



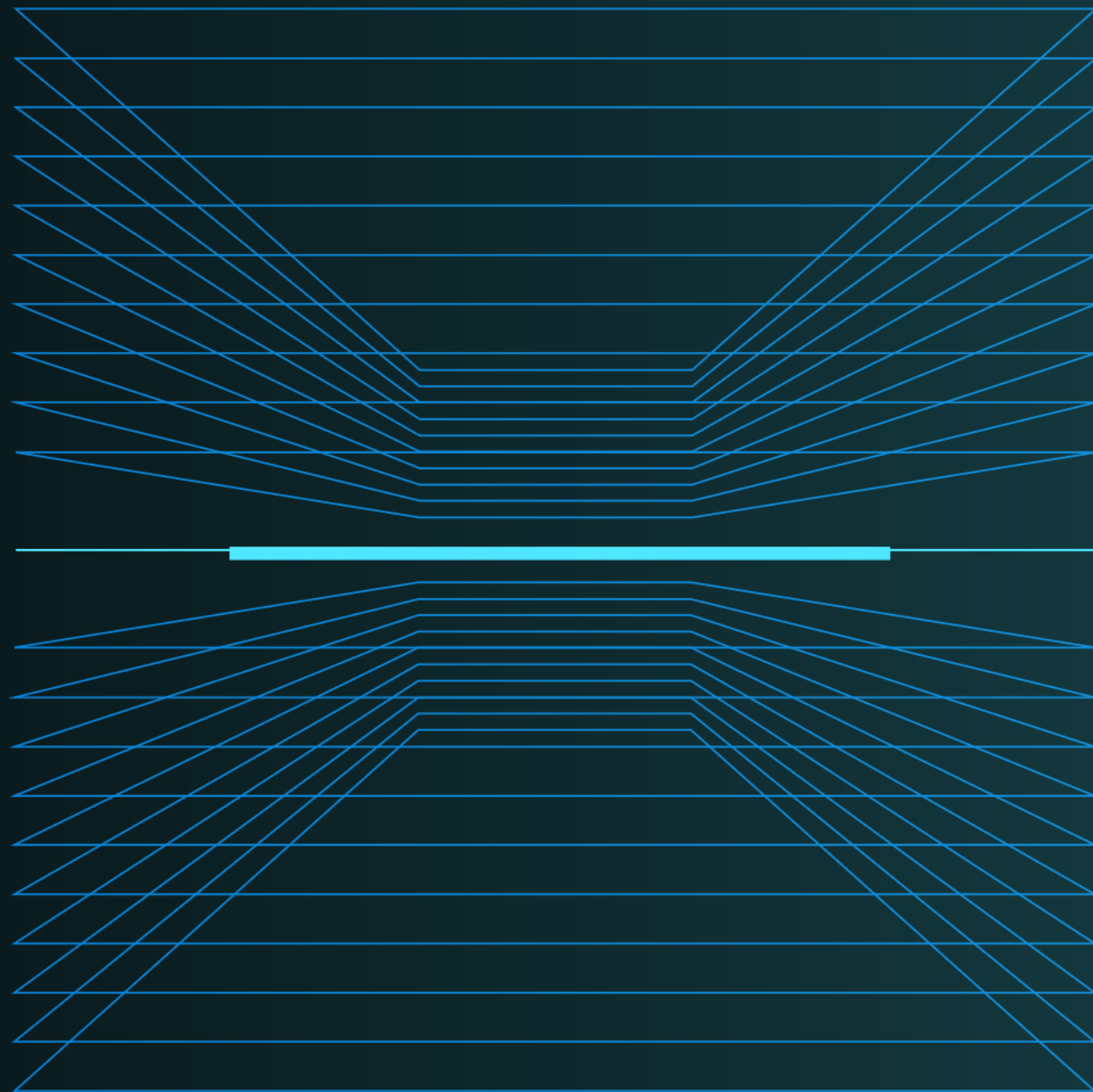
Azure OpenAI Service

A day in the life of a Data Scientist

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April 08, 2024



Learning objectives

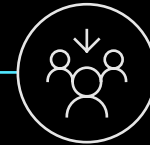
This set of slides will help understanding of:

How can Data Scientists/ Cloud Solution Architects leverage Azure OpenAI Service to build PoCs

A business use case definition to build an enterprise document search and Passage based QnA

Implementation on small set of documents (NLP Research Papers), playing with the Azure OpenAI studio, and leveraging the SDK

How to leverage Azure Form Recognizer, Azure OpenAI models, and Azure Cache for Redis Enterprise



Enterprise document search and Passage QnA

Current state

- Pharmaceutical companies have a large corpus of documents with information embedded on drug discovery process and historical experimentation
- In their drug discovery process, Scientists read through these documents to find details on historical experimentation
- Missing an end-to-end solution that enables the scientists to extract relevant information based on their query
- The process is time-consuming and requires manual labor—reducing the productivity of Scientists and slowing the drug discovery process

Final state

- An AI powered end-to-end search solution that could be leveraged to extract meaningful information from the document corpus based on the user query and provides crisp summarized results
- A solution that expedites the drug discovery process
- Shortens reading time for Scientists and increases bandwidth for humans to focus on more in-depth analysis
- Improves the effectiveness of searching for information

PoC on small set of NLP Research papers (in pdf)

For PoC, Enterprise document search and Passage QnA solution is implemented for a small set of documents.

The goal is to help user ask question related to a document and the solution will **provide most similar chunks of text as Passage** and enable the user to ask **specific question from passage**.

Benefits of the solution when built Enterprise grade:

- Less number of token transactions with Azure OpenAI models
- Passage based on question answer
- May reduce latency

List of papers used for demonstration

- Attention is all you need
- BART
- BERT
- GPT3
- InstructGPT
- Sparrow

Azure OpenAI Playground demo

Enabling all skill levels to build prototypes easily and explore the art of possible

Cognitive Services | Azure OpenAI Studio - Preview



Chris Hoder
BugBashTest6 (South Central US, S0)

CH

Azure OpenAI Studio

Privacy & cookies

Get started with Azure OpenAI

Perform a wide variety of natural language tasks with Azure OpenAI, including copywriting, summarization, parsing unstructured text, classification, and translation.



Experiment with prompt completions

Try out the completions endpoint by writing a prompt and generating a response. Set different parameters values to adjust how the model responds.

[Go to playground](#)



Customize a model with fine-tuning

Fine-tune a custom model to increase reliability for a wide variety of use cases while decreasing costs and speeding up processing times.

[Start fine-tuning a custom model](#)



Manage deployments in your resource

Create deployments to explore the model capabilities.

[Go to Files](#)



Learn more with documentation

Dive right in with quickstart guides, or browse technical specifications for the Azure OpenAI APIs.

[Explore documentation](#)

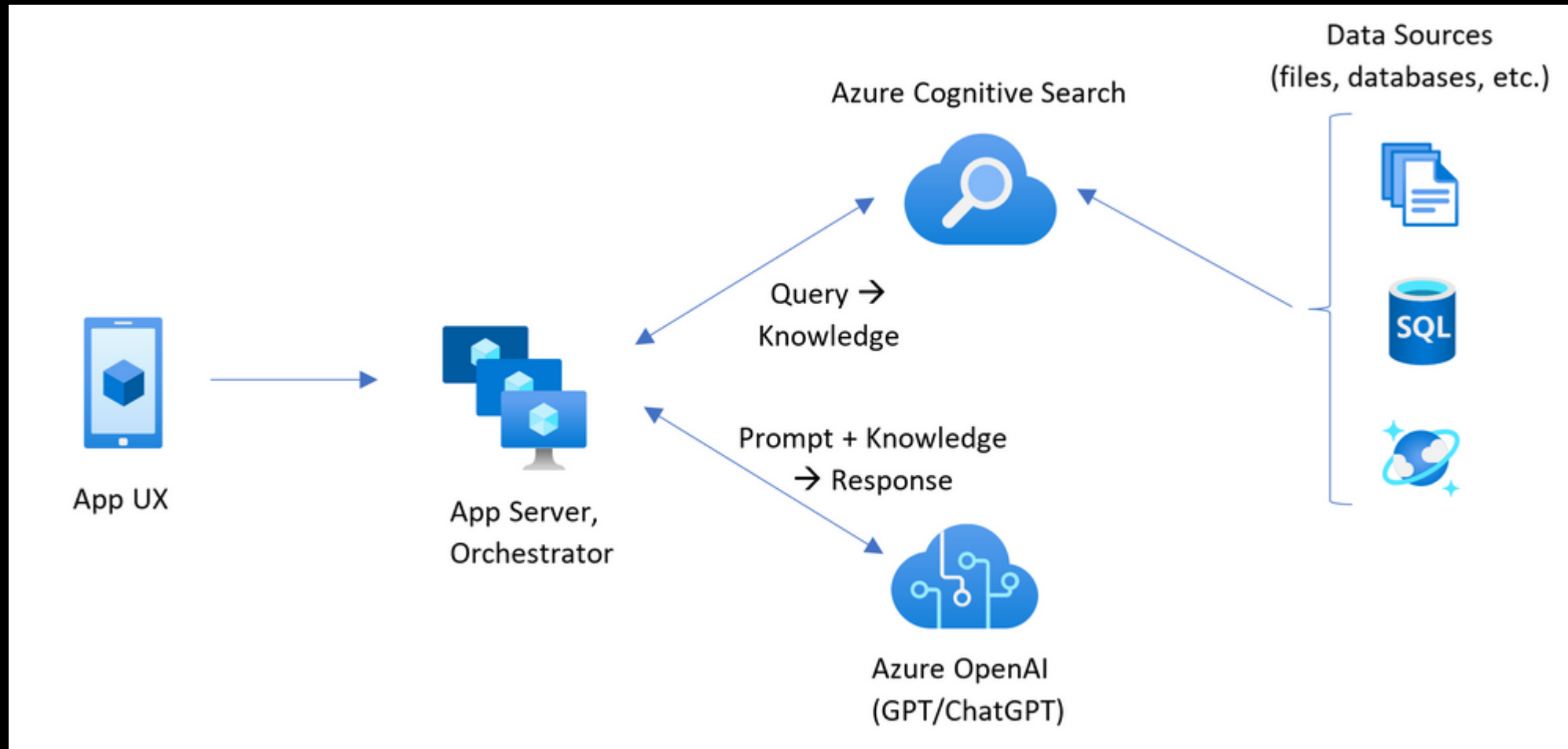
Explore examples for prompt completion

[Featured examples](#) [View all](#) [Q&A](#) [Chat with an assistant](#) [Summarization](#) [Translation](#) [Extraction](#)

Azure OpenAI Playground demo

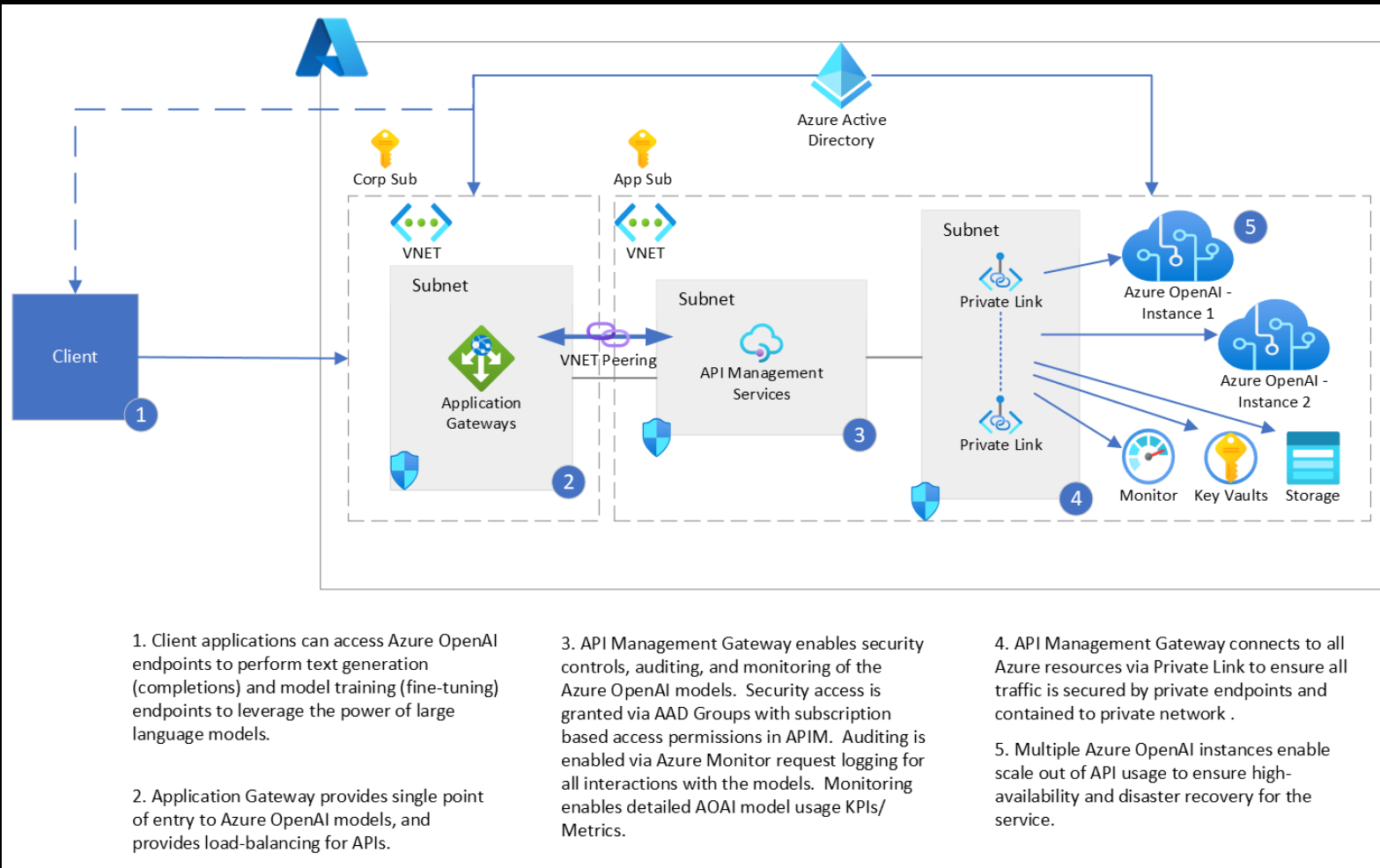
ChatGPT + Enterprise data with Azure OpenAI and Cognitive Search

[GitHub - Azure-Samples/azure-search-openai-demo](#): A sample app for the Retrieval-Augmented Generation pattern running in Azure, using Azure Cognitive Search for retrieval and Azure OpenAI large language models to power ChatGPT-style and Q&A experiences.



Azure OpenAI Playground demo

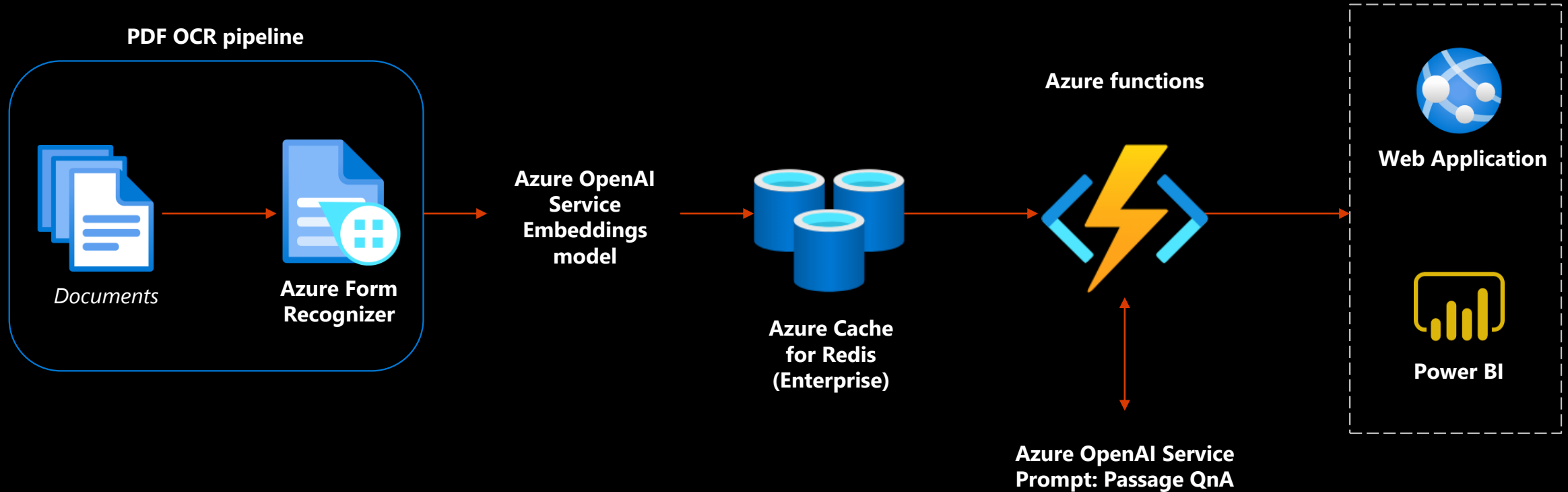
Enterprise Azure OpenAI



<https://github.com/Azure-Samples/openai-python-enterprise-logging>

[Azure-Samples/azure-search-openai-demo: A sample app for the Retrieval-Augmented Generation pattern running in Azure, using Azure Cognitive Search for retrieval and Azure OpenAI large language models to power ChatGPT-style and Q&A experiences. \(github.com\)](#)

Architecture design to build the PoC



Step-by-step process flow in the Jupyter Notebook



Step-by-step process flow

- Use Form Recognizer to extract text from NLP papers
- Chunk the text and create embeddings using Azure OpenAI embeddings model
- Spin up Azure Cache for Redis (Enterprise) and store the embeddings
- Leverage vector Similarity feature of Cache for Redis to get top n similar chunks
- Take top n similar texts as a passage and prompt engineer into passage based QnA solution

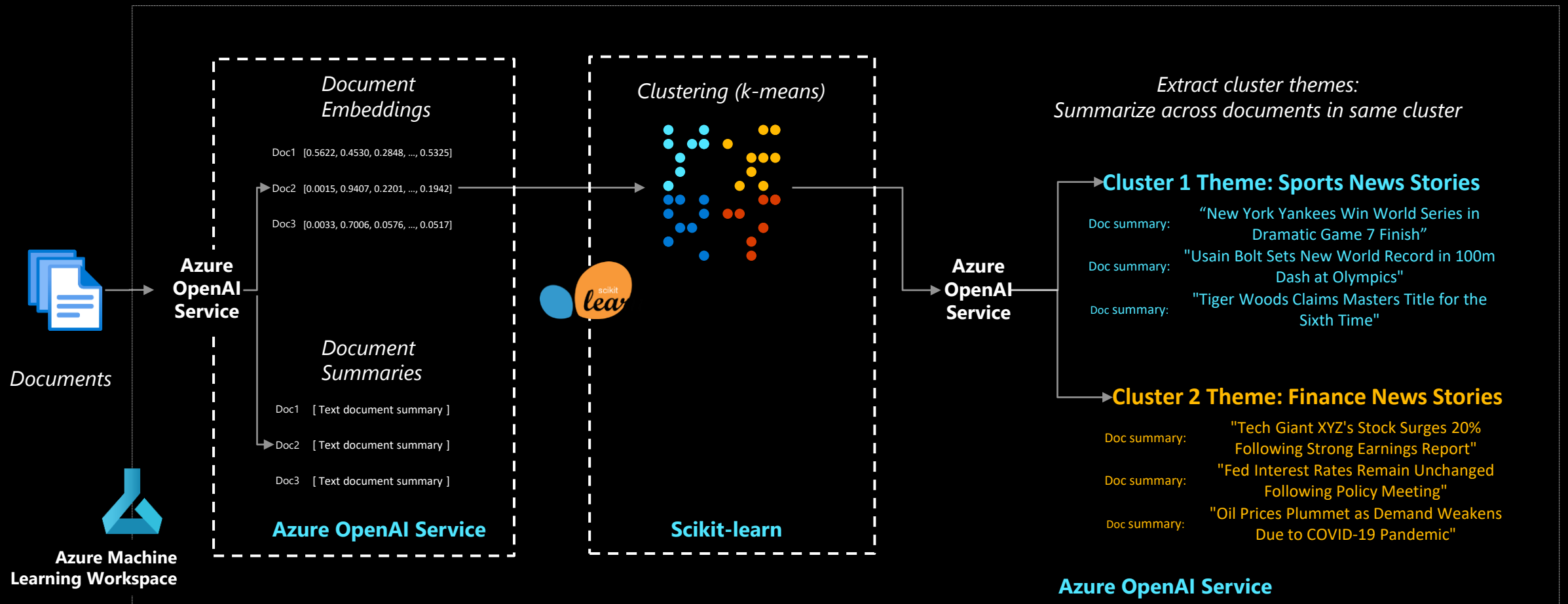
GitHub Repo: <will clean code, add to the GitHub repo, provide the link here>

Considerations

- Slides comprise of a business use case and demo implementation (PoC)
- PoC implementation is not an enterprise grade
- For this PoC, Azure Redis for Cache Enterprise version is used which is expensive
- This PoC is devised for Enterprise Grade search and QnA scenarios
- We can use any embeddings models based on the use case
- Prompt engineering is the key
(Prompt engineering samples provided in the deck)
- Playing with the parameters like temperature can help customize the result based on the use case
- Fine-tuning the models could also help

Document Clustering, and Cluster Theme Generation

Summarize and extract document similarity embeddings,
mine for similar clusters, generate cluster themes



Lease processing with zero-shot learning approach

Send a single prompt to model

The prompt consists of the:
Natural language instruction task,
in this case, along with the text, or
lease document

Azure OpenAI model extracts specific entities

With no additional training, or label-dataset,
the model extracts specific entities based on
the natural language instruction and
document context

...and returns in the format requested

Custom named entity recognition requires fine-tuning a large model, with your own, labeled data samples. Here, the model extracted the custom entities with 0 examples or fine-tuning!

Using the lease below return a json object containing the landlord, tenant, address of leased property, agreement date, start of the contract, etc..

This House Rental Agreement ("Agreement", "rental agreement," or "lease") is entered between Opay LLC (Landlord) and Sam Zamora and Garry Byrne and Sarah Cantrel (tenants). If more than one person is named as Tentant they shall be....



HOUSE RENTAL AGREEMENT

This House Rental Agreement ("Agreement," "rental agreement," or "lease") is entered into between Opay LLC (Landlord) and Sam Zamora and Garry Byrne and Sarah Cantrel (Tenants). If more than one person is named as Tenant they shall be jointly and severally liable and responsible under the terms of this Agreement. This lease Agreement involves a residential house, yard, and related facilities located at 14 Bellevue Dr Salt Lake City Utah 84118 (the "premises"). The date of this Agreement is January 15 2024.

1. Landlord rents to Tenant, unfurnished, the premises on a month to month basis, terminable by either party at the end of any calendar month on at least 30 days notice to the other party. Tenant shall be entitled to possession of the premises and rent shall commence on April 15 2024. Tenant shall not assign, sublease, or allow anyone other than persons permitted under this lease to at any time be in possession of any portion of the premises. Landlord will provide five (5) keys to Tenant; each key fits all outside door locks. Landlord will also provide two keys to the garage, and one remote garage door opener. All keys and the remote door opener will be returned to Landlord at the end of the tenancy.

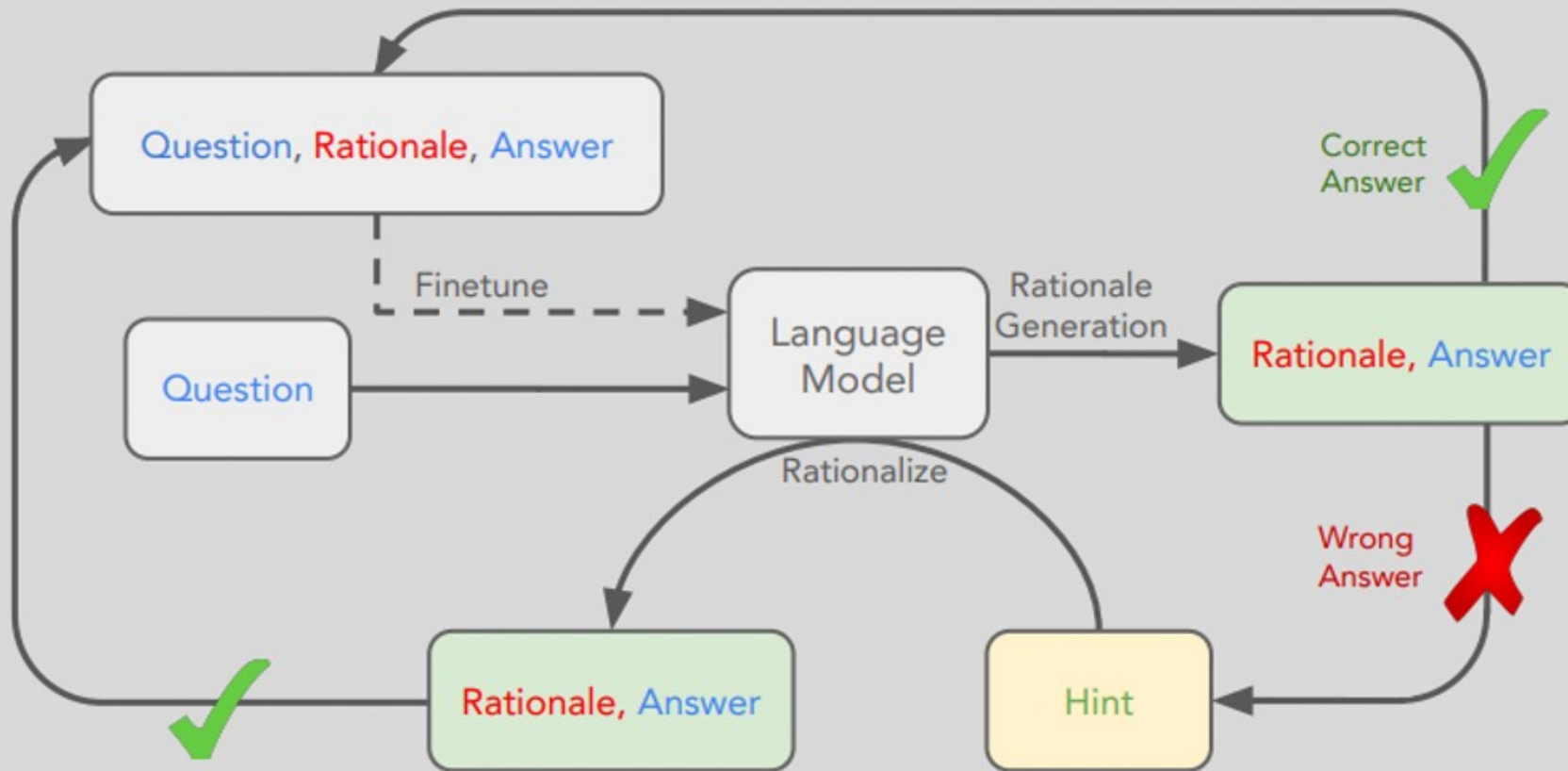
2. Tenant agrees to pay the stipulated monthly rent in advance on the first day of each calendar month without offset or reduction. All rent is payable to Landlord at the address of Landlord, 541 Old Terrace Road Salt Lake City Utah 84198 or at such other place as Landlord shall designate. Landlord's telephone number is (801) 568-5985.

3. The stipulated rent under this Agreement shall be prorated on a daily basis for any partial month, and calculated at the rate of \$2500 per month. Landlord agrees to give Tenant written notice 30 days in advance of any change in the rental amount. All rent or other amounts



```
{
  "Lessor": "Bob Marshall",
  "Lessee": "The United States of America",
  "Property Address": "#205 Lifton Avenue, Ft. Overview Heights, St. James, Barbados",
  "Property Description": "3 bedrooms, 2 bathrooms, living-dining room, family room, and kitchen",
  "Net Size of Property": 1725,
  "Net Size of Property Units": "square feet",
  "Gross Size of Property": null,
  "Gross Size of Property Units": null,
  "Contract Start Date": "1997-05-23",
  "Lease Start Date": "1997-06-28",
  "Lease End Date": "2003-06-27",
  "Lease Term": "6 years",
  "Monthly Lease Amount": "BDS$5,300",
  "Payment Frequency": "quarterly in advance",
  "Payment Address": "11 Parma Avenue, Ft. Overview Heights, St. James, Barbados"
}
```

Fine-Tuning with Chain-of-Thought



How to adapt GPT-3 model for your task

No Gradient Updates

Zero-Shot

The model predicts the answer given only a natural language description of the task.

One-Shot

In addition to the task description, the model sees a single example of the task

Few-Shot

In addition to the task description, the model sees a few examples of the task.

Fine Tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.

Prepare and upload
training data



Train a new fined
tuned model



Use your fine-tuned
model

1.

Higher quality results
than prompt design

2.

Ability to train on more examples
than can fit in a prompt

3.

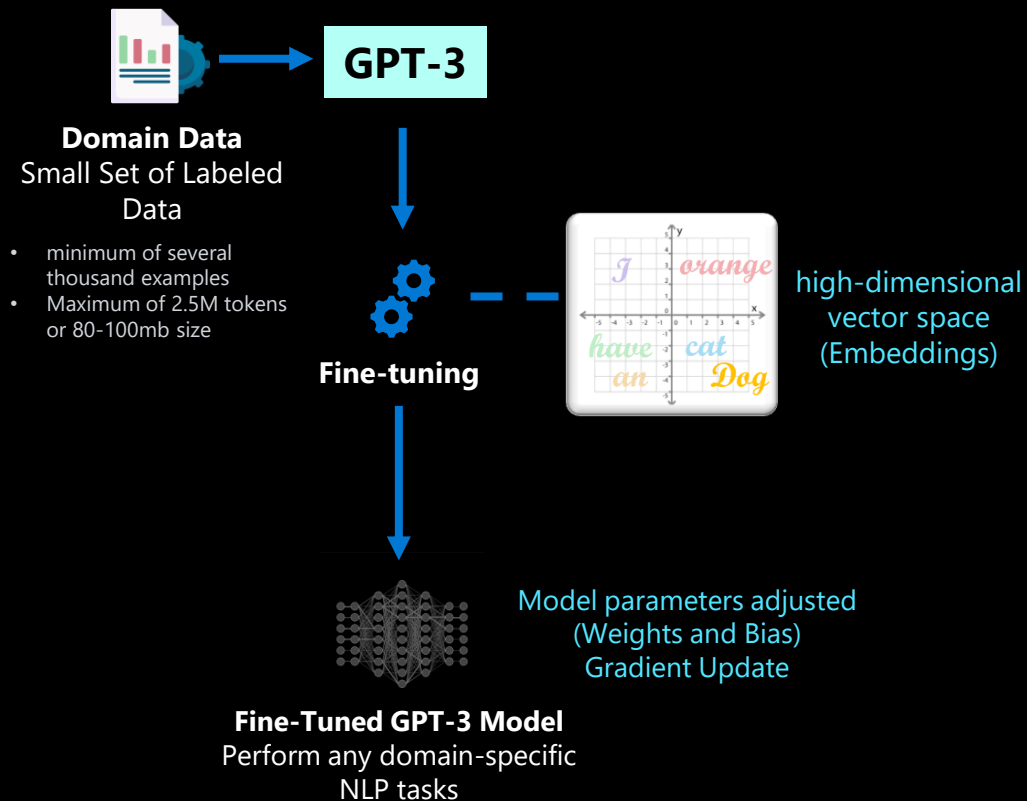
Token savings due
to shorter prompts

4.

Lower latency requests

Model Adaptation with specific domain data

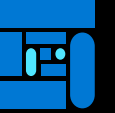
Fine-Tuning



Fine-tuning results is a new model being generated with updated weights and biases.

This is in contrast to **few-shot learning** in which model weights and biases are not updated.

When Fine-Tuning is needed



If model is making untrue statements ("hallucinations"), then mitigate the hallucinations
Accuracy of results of the model does not meet customer requirements

Fine-tuning lets you get more out of the models available through the API by providing:

- Higher quality results than prompt design
- Ability to train on more examples than can fit in a prompt
- Lower latency requests

Fine-tuning improves over few-shot learning by training on many more examples than can fit in the prompt, letting you achieve better results on a wide number of tasks.

Best practices of Fine-Tuning



Fine-tuning data set must be in JSON format

A set of training examples that each consist of a single input ("prompt") and its associated output ("completion")

For classification task, the prompt is the problem statement, completion is the target class

For text generation task, the prompt is the instruction/question/request, and completion is the text ground truth

Best practices of Fine-Tuning



Fine-tuning data size: Advanced model (Davinci) performs better with limited amount of data; with enough data, all models do well.

Fine-tuning performs better with more high-quality examples.

To fine-tune a model that performs better than using a high-quality prompt with base models, you should provide at least a few hundred high-quality examples, ideally vetted by human experts.

From there, performance tends to linearly increase with every doubling of the number of examples. Increasing the number of examples is usually the best and most reliable way of improving accuracy.

Best practices of Fine-Tuning



Make sure to completely remove wrong labels in fine-tuning dataset. If you are fine-tuning on a pre-existing dataset rather than writing prompts from scratch, be sure to manually review your data for offensive or inaccurate content if possible, or review as many random samples of the dataset as possible if it is large.

Fine-Tuning data formatting



To fine-tune a model, you'll need a set of training examples that each consist of a single input ("prompt") and its associated output ("completion").

This is notably different from using the base models, where you might input detailed instructions or multiple examples in a single prompt.

Each prompt should end with a fixed separator to inform the model when the prompt ends and the completion begins.

A simple separator which generally works well is `\n\n###\n\n`.

The separator should not appear elsewhere in any prompt.

Fine-Tuning Data Formatting



Each completion should start with a whitespace due to tokenization, which tokenizes most words with a preceding whitespace.

Each completion should end with a fixed stop sequence to inform the model when the completion ends.

A stop sequence could be `\n`, `###`, or any other token that does not appear in any completion.

For inference, you should format your prompts in the same way as you did when creating the training dataset, including the same separator.

Also specify the same stop sequence to properly truncate the completion.

Hyperparameters specific to Fine-Tuning

Parameter	Description	Recommendation
n_epochs controls how many times each example is trained on	The number of epochs to train the model for. An epoch refers to one full cycle through the training dataset.	Start from 4 and small datasets may need more epochs and large datasets may need fewer epochs. If you see low training accuracy (underfitting), try increasing n_epochs. If you see high training accuracy but low validation accuracy (overfitting), try lowering n_epochs.
batch_size controls the number of training examples used in a single training pass	The batch size to use for training. The batch size is the number of training examples used to train a single forward and backward pass.	We've seen good performance in the range of 0.01% to 4% of training set size. In general, we've found that larger batch sizes tend to work better for larger datasets.
learning_rate_multiplier controls rate at which the model weights are updated	The learning rate multiplier to use for training. The fine-tuning learning rate is the original learning rate used for pre-training, multiplied by this value.	We recommend experimenting with values in the range 0.02 to 0.2 to see what produces the best results. Empirically, we've found that larger learning rates often perform better with larger batch sizes. Empirically, we found learning_rate_multiplier has minor impact compared to n_epochs and batch_size.
prompt_loss_weight controls how much the model learns from prompt tokens vs completion tokens	The weight to use for loss on the prompt tokens. This value controls how much the model tries to learn to generate the prompt (as compared to the completion, which always has a weight of 1.0.) Increasing this value can add a stabilizing effect to training when completions are short.	When a model is fine-tuned, it learns to produce text it sees in both the prompt and the completion. In fact, from the point of view of the model being fine-tuned, the distinction between prompt and completion is mostly arbitrary. The only difference between prompt text and completion text is that the model learns less from each prompt token than it does from each completion token. This ratio is controlled by the prompt_loss_weight, which by default is 0.1. If prompts are extremely long (relative to completions), it may make sense to reduce this weight to avoid over-prioritizing learning the prompt. Empirically, we found prompt_loss_weight has minor impact compared to n_epochs and batch_size.

Guides and examples for Fine-Tuning

Getting started with Fine-Tuning:

[How to customize a model with Azure OpenAI - Azure OpenAI | Microsoft Learn](#)

[Fine-tuning—OpenAI API](#)

<https://platform.openai.com/docs/guides/fine-tuning/general-best-practices>

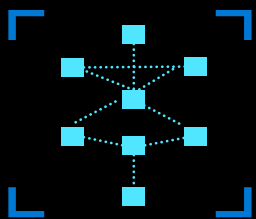
Detailed examples of fine-tuning

[\[PUBLIC\] Best practices for fine-tuning GPT-3 to classify text—Google Docs](#)

Using fine-tuning to improve truthfulness

[Fine-tuning a Classifier to Improve Truthfulness | OpenAI Help Center](#)

Embeddings



An embedding is a special format of data representation that can be easily utilized by machine learning models and algorithms.

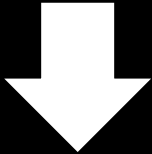
The embedding is an information dense representation of the semantic meaning of a piece of text.

Each embedding is a vector of floating-point numbers, such that the distance between two embeddings in the vector space is correlated with semantic similarity between two inputs in the original format.

For example, if two texts are similar, then their vector representations should also be similar.

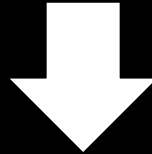
Embeddings make it possible to map content to a “semantic space”

A neutron star is the collapsed core of a massive supergiant star



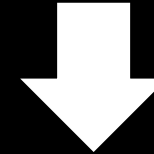
[15 34 24 13 ...]

A star shines for most of its active life due to thermonuclear fusion.



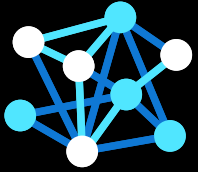
[16 22 89 26 ...]

The presence of a black hole can be inferred through its interaction with other matter



[20 13 31 89 ...]

Embedding models



Different Azure OpenAI embedding models are specifically created to be good at a particular task.

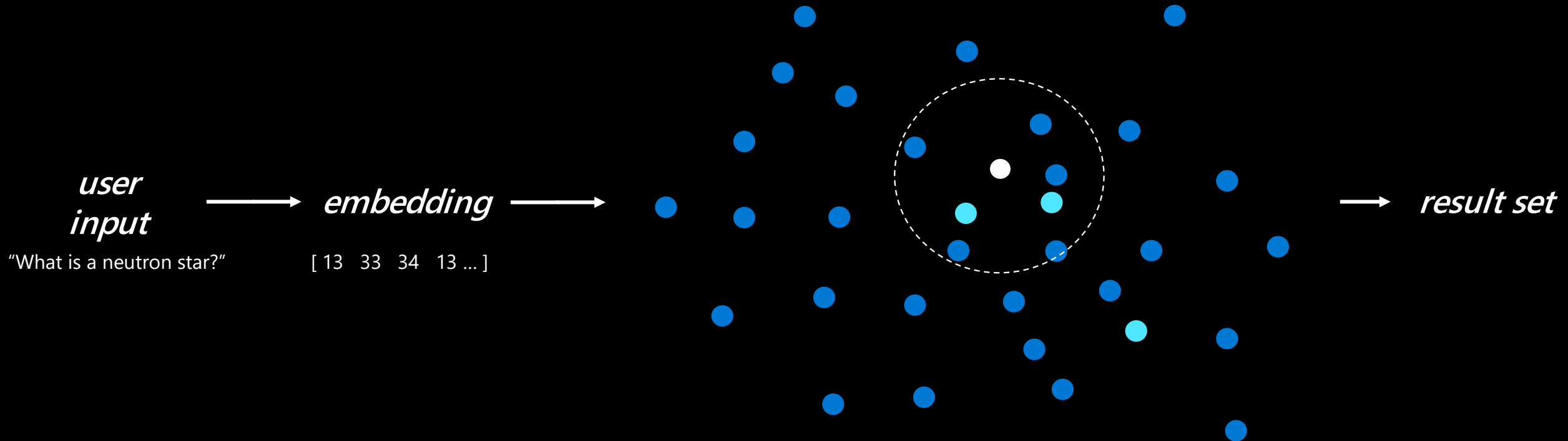
- **Similarity embeddings** are good at capturing semantic similarity between two or more pieces of text.
- **Text search embeddings** help measure long documents are relevant to a short query.
- **Code search embeddings** are useful for embedding code snippets and embedding nature language search queries.

Embeddings make it easier to do machine learning on large inputs representing words by capturing the semantic similarities in a vector space.

Therefore, we can use embeddings to determine if two text chunks are semantically related or similar, and provide a score to assess similarity.

Similarity Search with embeddings

Once you encode your content as embeddings, you can then get an embedding from the user input and use that to find the most semantically similar content.



Industry examples and uses for GPT-3

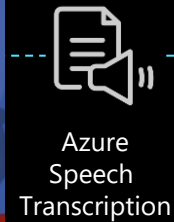
Customer Service	GPT-3 is used to automate response to customer inquiries
Reporting	Generate reports with GPT-3, making it easier for companies to get the most up-to-date information
Knowledge & Information	GPT-3 helps organizations track knowledge and information, making it easier for employees to find and use information they need
Content Generation	GPT-3 automatically generates content on various topics
Sentiment Analysis	GPT-3 analyzes the sentiment of text data to understand customer sentiment and make better decisions
Natural Language Processing	GPT-3 processes and understand natural language data, making it easier to extract information from text data
Machine Translation	GPT-3 automatically translates text from one language to another
Predictive Modeling	GPT-3 builds predictive models that can be used to make better decisions about the future.
Analytics	GPT-3 can extract insights from data that facilitate informed decision making about their business
Security	GPT-3 can protect data and prevent unauthorized access

Media example

News Analyses & Article Creations



News Broadcast
(Global Warming)



Azure
OpenAI Service
Text Completion

Search
SEO
Virtual Agent
Analytics
Reporting
Knowledge Mining

Content Analysis

Content Creation

Ideation
Productivity
Personalization
Accessibility
A/B Testing

Topic Classification

Global warming, Deforestation, Carbon footprint

Entity Extraction

Organizations: IPCC, UNFCCC, Green Peace
Geography: Canada, USA

Key Word Extraction

Human activities, fossil fuels, earth atmosphere

Question and Answer

What is the Intergovernmental Panel on Climate Change (IPCC)?
The IPCC is an international organization that studies climate change and the effectiveness ...

Video summarization

The article discusses about global warming and its effects on the Earth's atmosphere, wildlife, and human communities. It states that the primary cause of global warming is

News article generation (or blogs, social media)

Global warming is the gradual increase in the overall temperature of the Earth's atmosphere, primarily caused ...

Script Generation

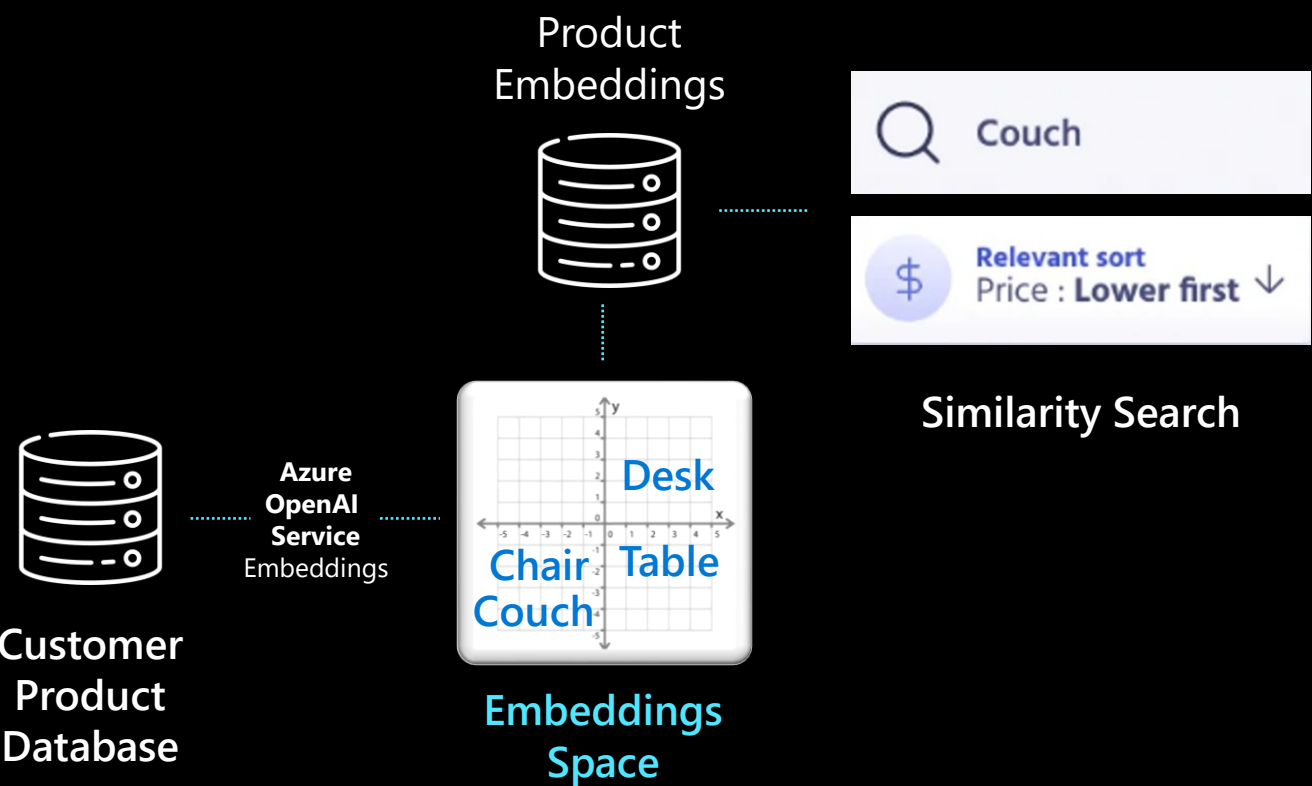
Act 1: The show opens with a shot of a beautiful coastal town
Act 2: As the town struggles to cope with the crisis ...

Personalized Content generation (or Advertising)

Simon, as someone passionate about global warming, you are aware of the urgent threat it poses to our planet ...

Retail example

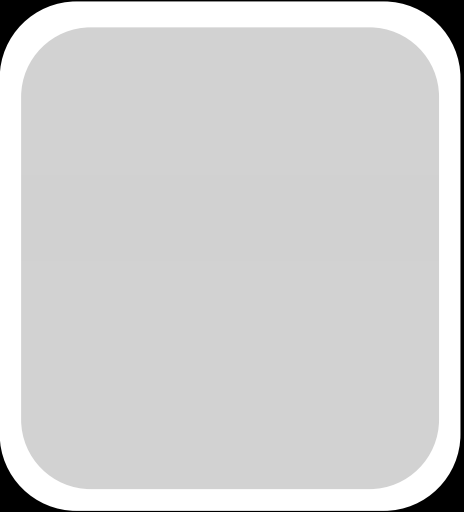
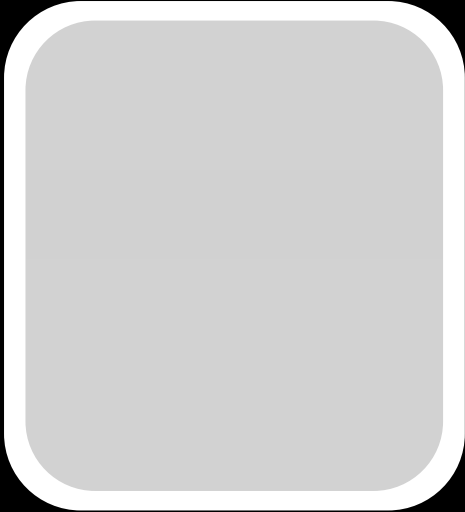
Similarity search



Saffron Ella \$2,300
Signature series

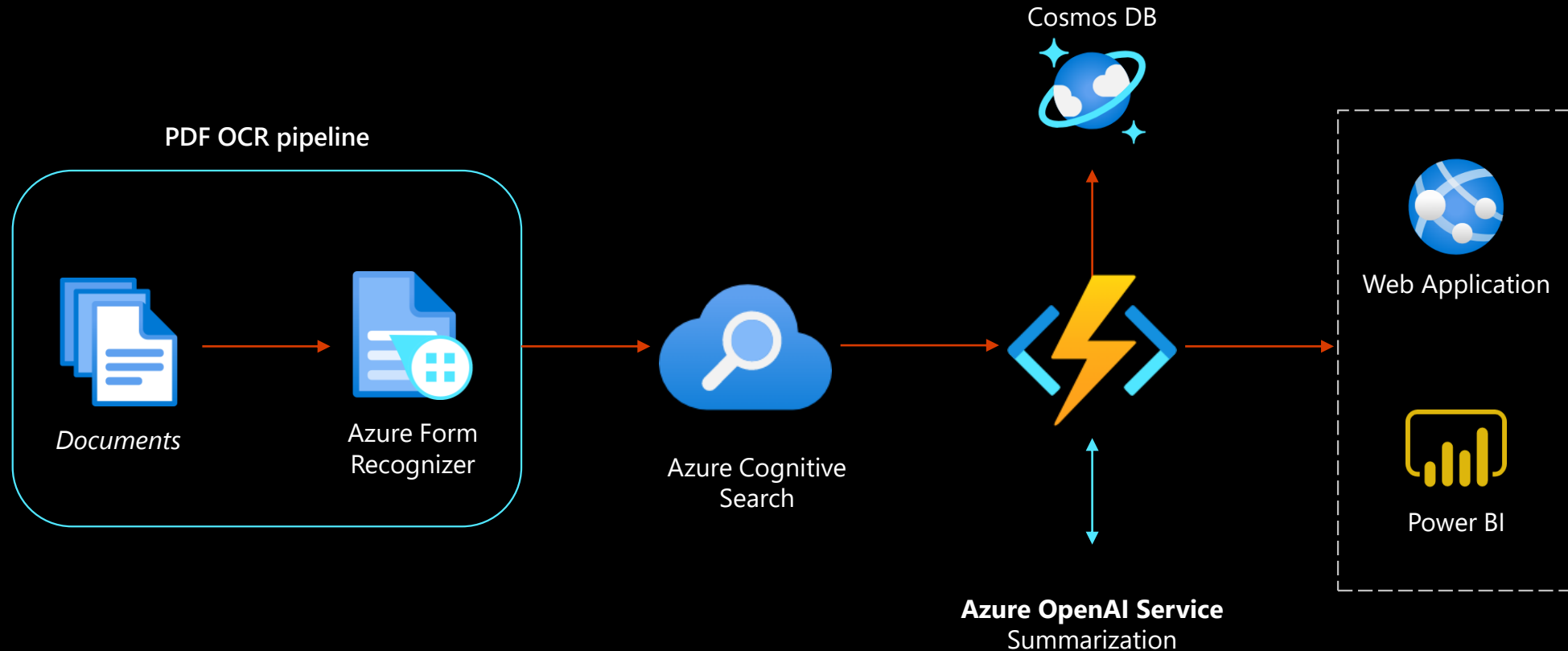


Slate Ray \$2,300
Signature series



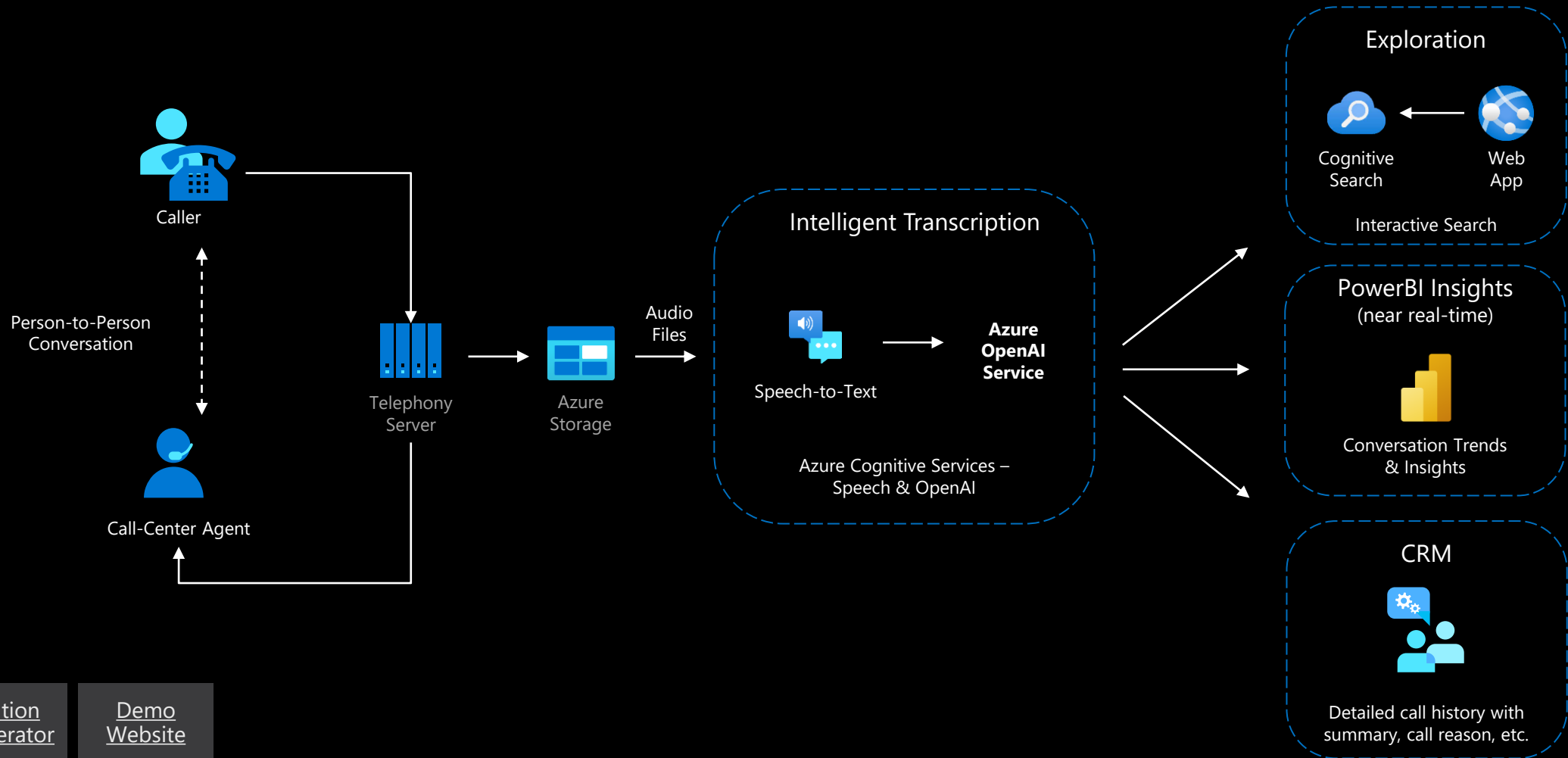
Azure OpenAI | GPT-3 sample high-level architecture

Document processing and summarization

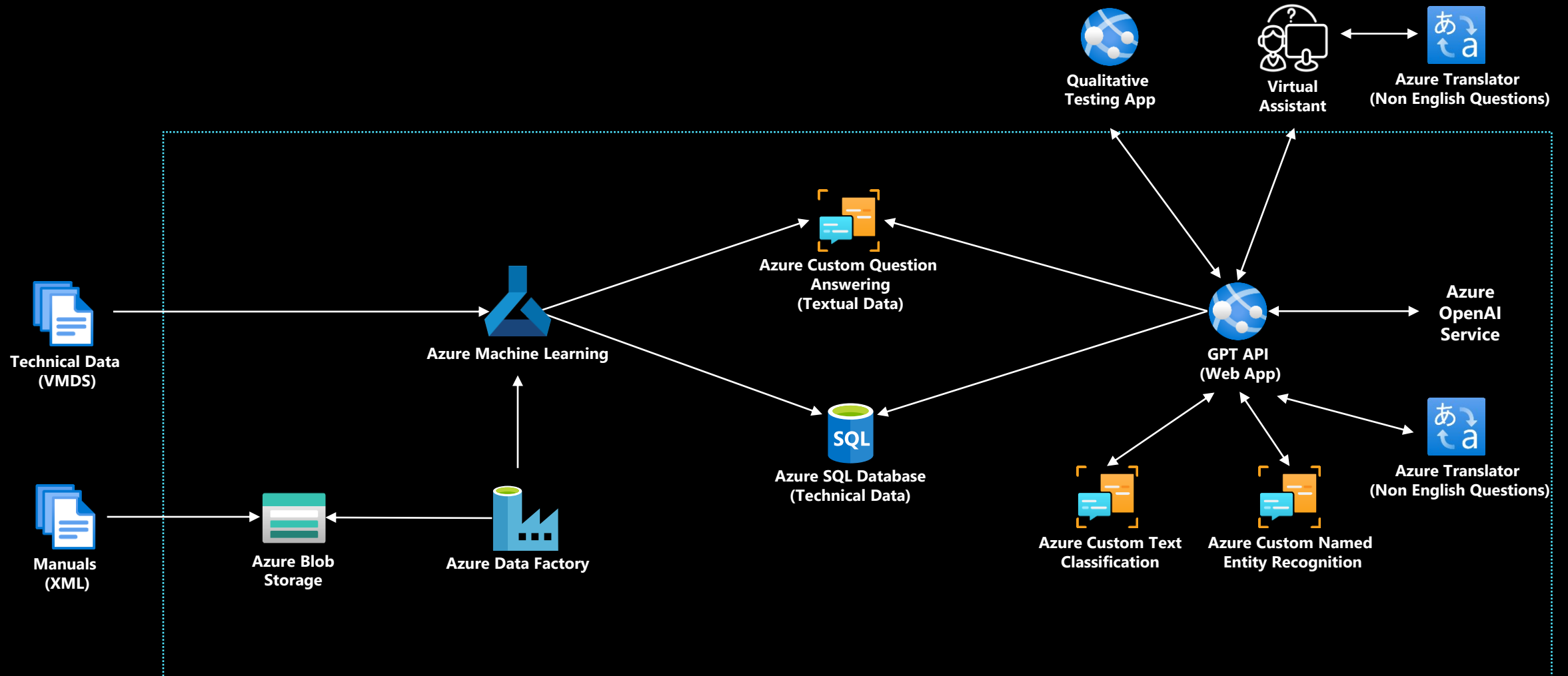


Contact Center Analytics using Speech API & OpenAI

Extract rich insights from call transcripts



Azure OpenAI Virtual Assistant architecture example



Component descriptions

Azure Blob Storage

- Stores unprocessed and processed data
- The update of data triggers Azure Data Factory

Azure Data Factory

- Triggering Data Pipelines

Azure Machine Learning

- AML Notebooks to pre-process the data and save the outcomes to the SQL DB and Azure Custom Question Answering

Longtail Search API

- Provides the API interface to the Longtail Search solution for the Virtual Assistant
- Modular Architecture for scalability with application structure
- Can hand over metadata to the application

Azure SQL

- Stores pre-processed manuals data, API responses and user feedback
- Using in-memory tables for improved performance

Azure OpenAI Service

- Used to generate SQL statements and generative textual answers

Azure Custom Question Answering

- Contains pre-processed data (owner manuals and component marketing texts)
- Used to rank text chunks based on question
- Can return confidence score of ranking result

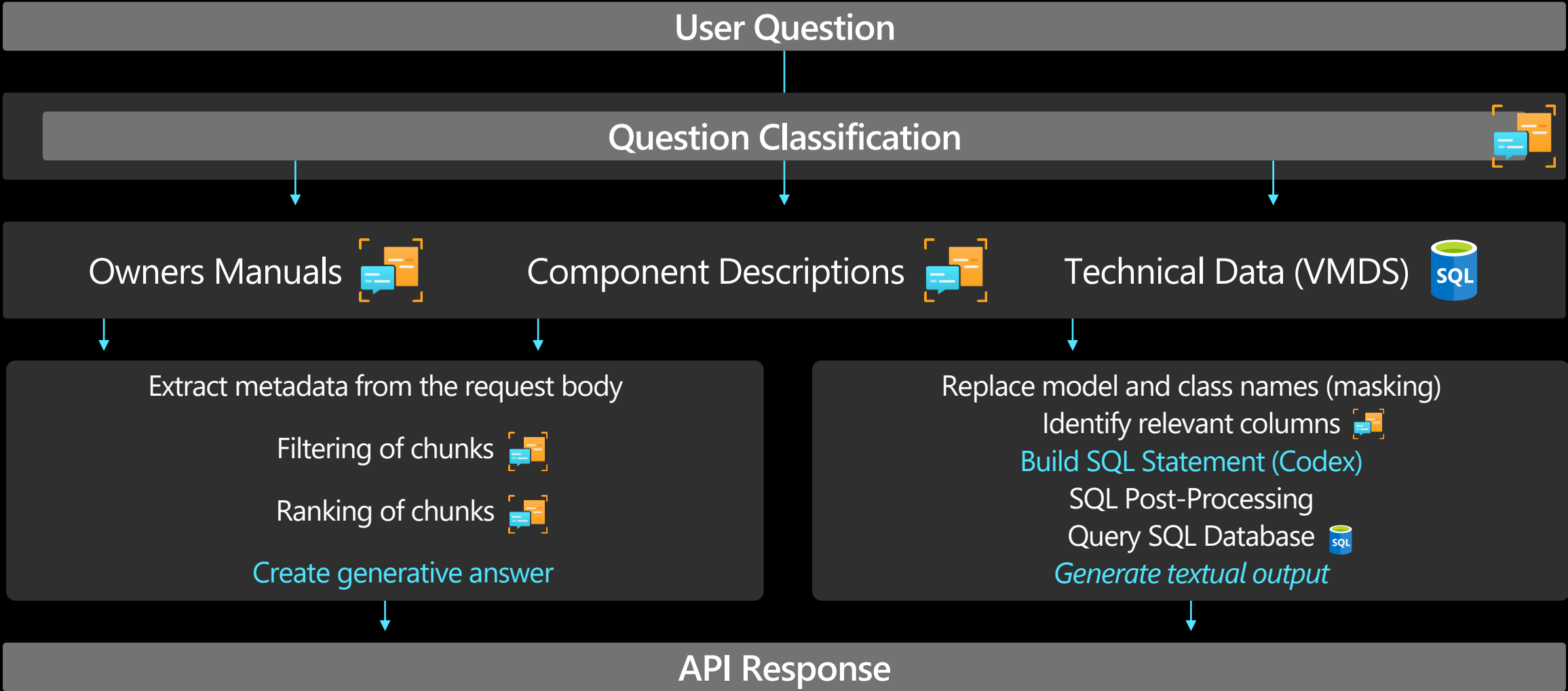
Azure Custom Named Entity Recognition

- Identify which columns are relevant to answer a user question.
- CNER is a classical NLP supervised learning tool. We trained and deployed a model with user questions which can identify the relevant columns

Azure Custom Text Classification

- To improve the quality of question-answering, we found out that the question classification adds a better focus. The user questions are classified into categories

Question answering process



Azure OpenAI Consumption Security Models



Data Access Controls



Anonymization and Pseudonymization



Encryption



Secure Data Transmission



Data Minimization



Regular Security Updates



Monitoring and Logging



Security Audits and Assessments

Azure OpenAI Service Responsible AI



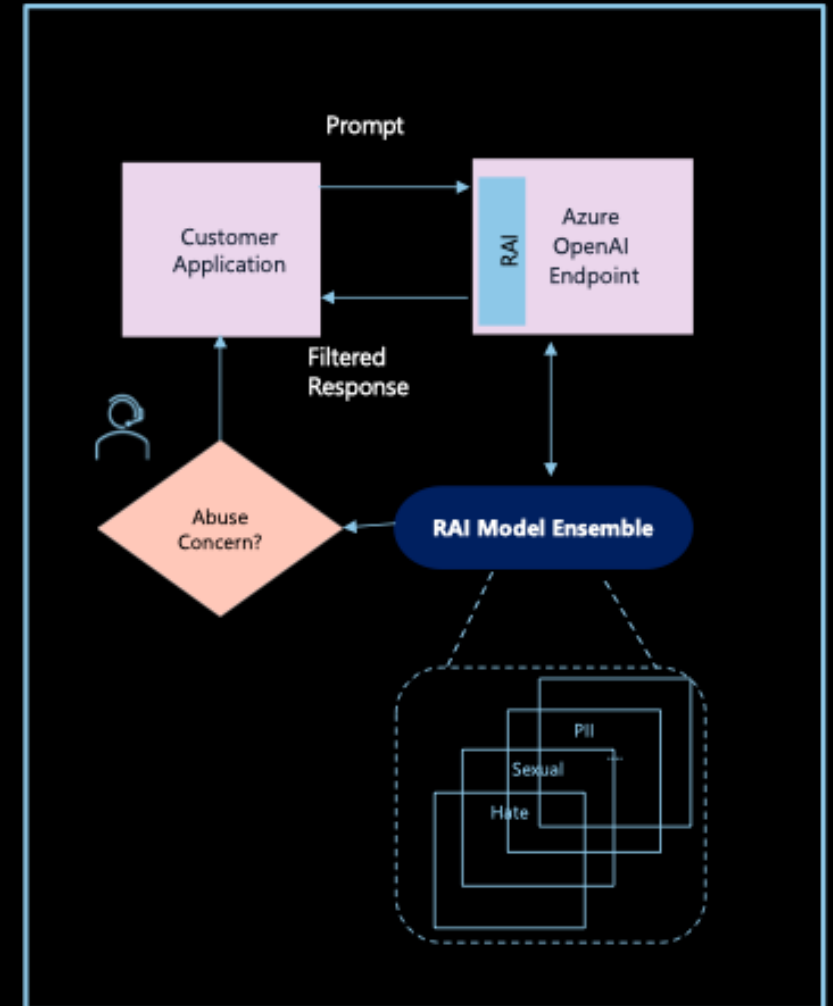
Custom content filters—
tailor tone and topics to your application



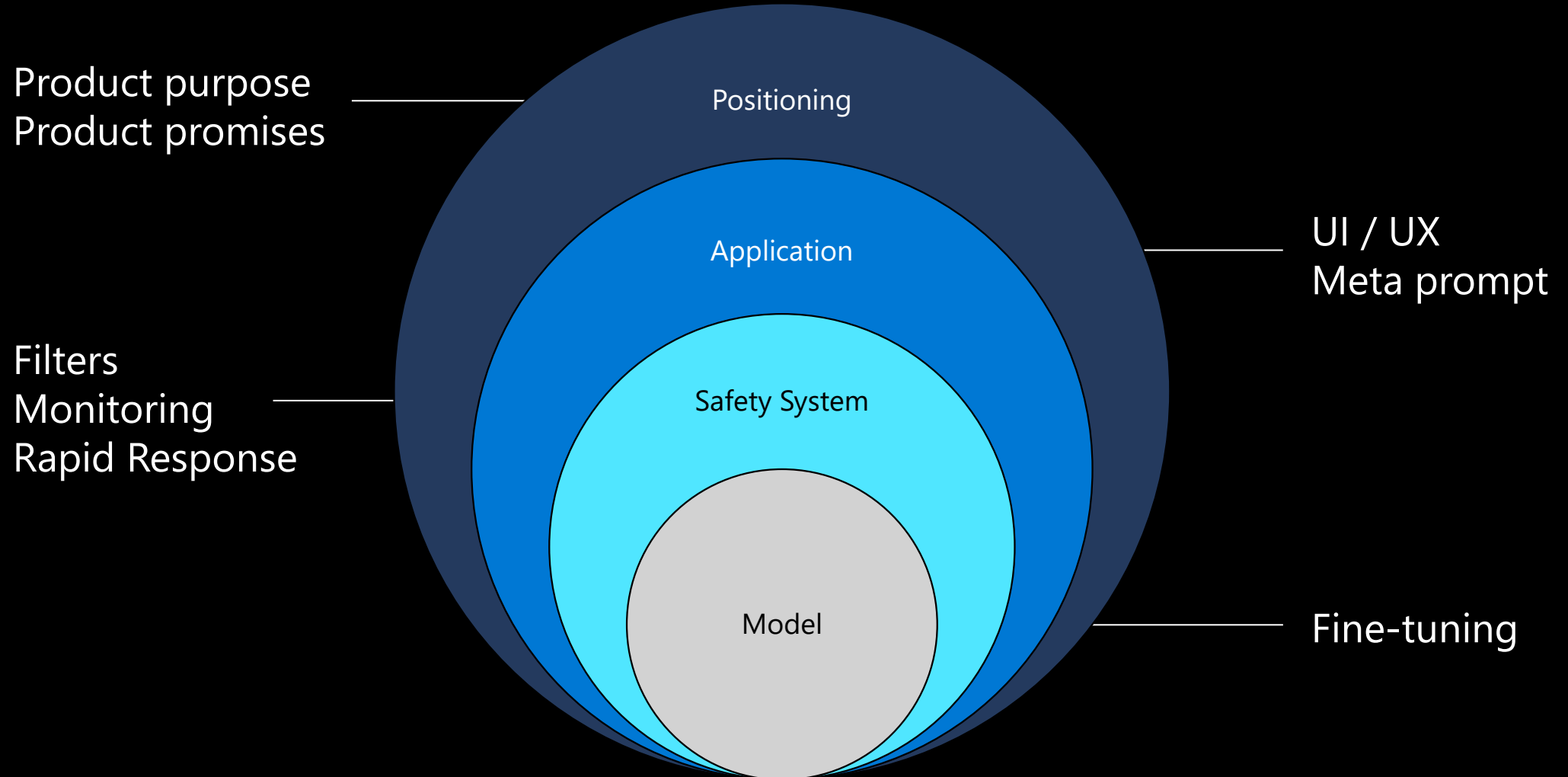
Abuse detection—
ensure responsible use of your application



Implementation guidelines, patterns,
and best practices



Mitigation Layers



RAI Mitigations

Customer	Structure user interactions. Limit the length, structure, and source of inputs and outputs
	Control user access
	Transparency and overreliance mitigations in UI/UX
Technical	Content Filtering
	Asynchronous abuse detection
	User-based throttling
	User-based shutdown
Process and Policy	Limited Access
	Abuse reporting channel
	Feedback channel
	Incident Response
Documentation and legal	Terms of use
	Transparency Note
	Design Guidelines

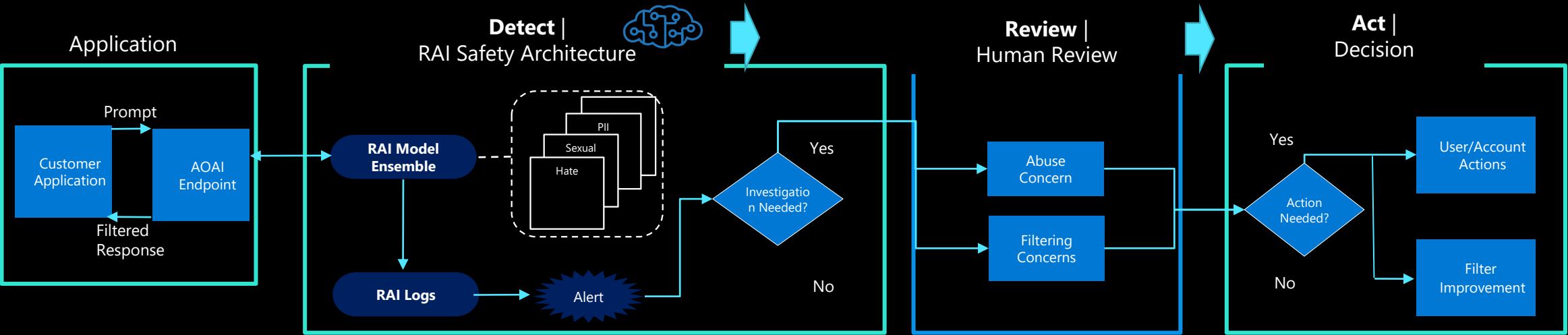
Quality

Harm to individuals or businesses due to unintended outputs or overreliance

INCIDENT CLASS SHORTHAND	INCIDENT CLASS EXTENDED DESCRIPTION
Inaccurate Text	API generates misleading, inaccurate, or poorly-contextualized content on high-stakes topics
Incorrect/insecure code	API generates incorrect or insecure code that is used unknowingly by users
PII	API generates responses that contain email addresses, SSNs, and other PII that is
Proprietary Info/plagiarism	API generates based on content or code that is proprietary
Demeaning, stereotyping, hate	API generates content that is offensive toward members of social groups
Inequitable allocation	API outputs lead to inequitable allocation of resources (e.g., likelihood of receiving job interview based on automated resume screener)
Quality of service harms	API systematically performs worse on text by, for, and about different social groups
Violence or Self-Harm	API instructs, affirms, or radicalizes a human to commit direct harm to themselves or others
Profane, sexual, inappropriate, or sensitive content	API generates contextually inappropriate, offensive, or sensitive content

Safety Execution Workflow

 People & Policy



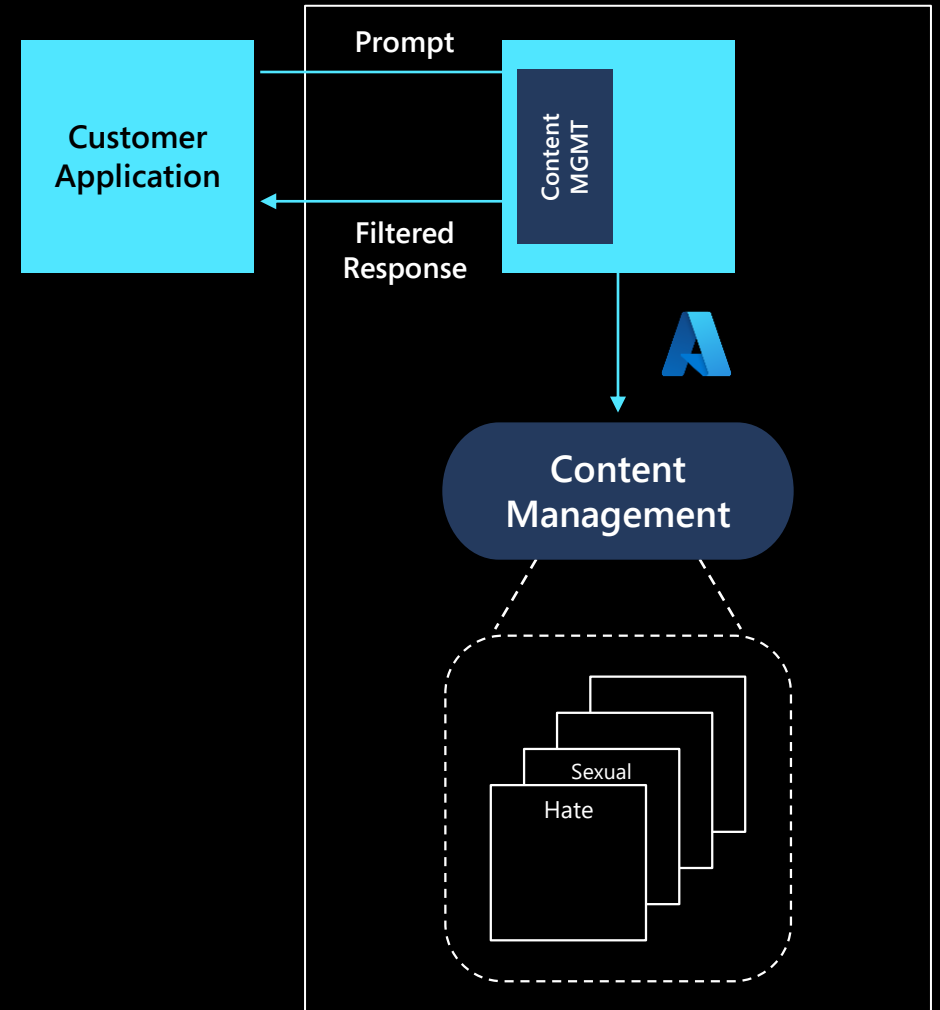
Azure OpenAI Service Responsible AI



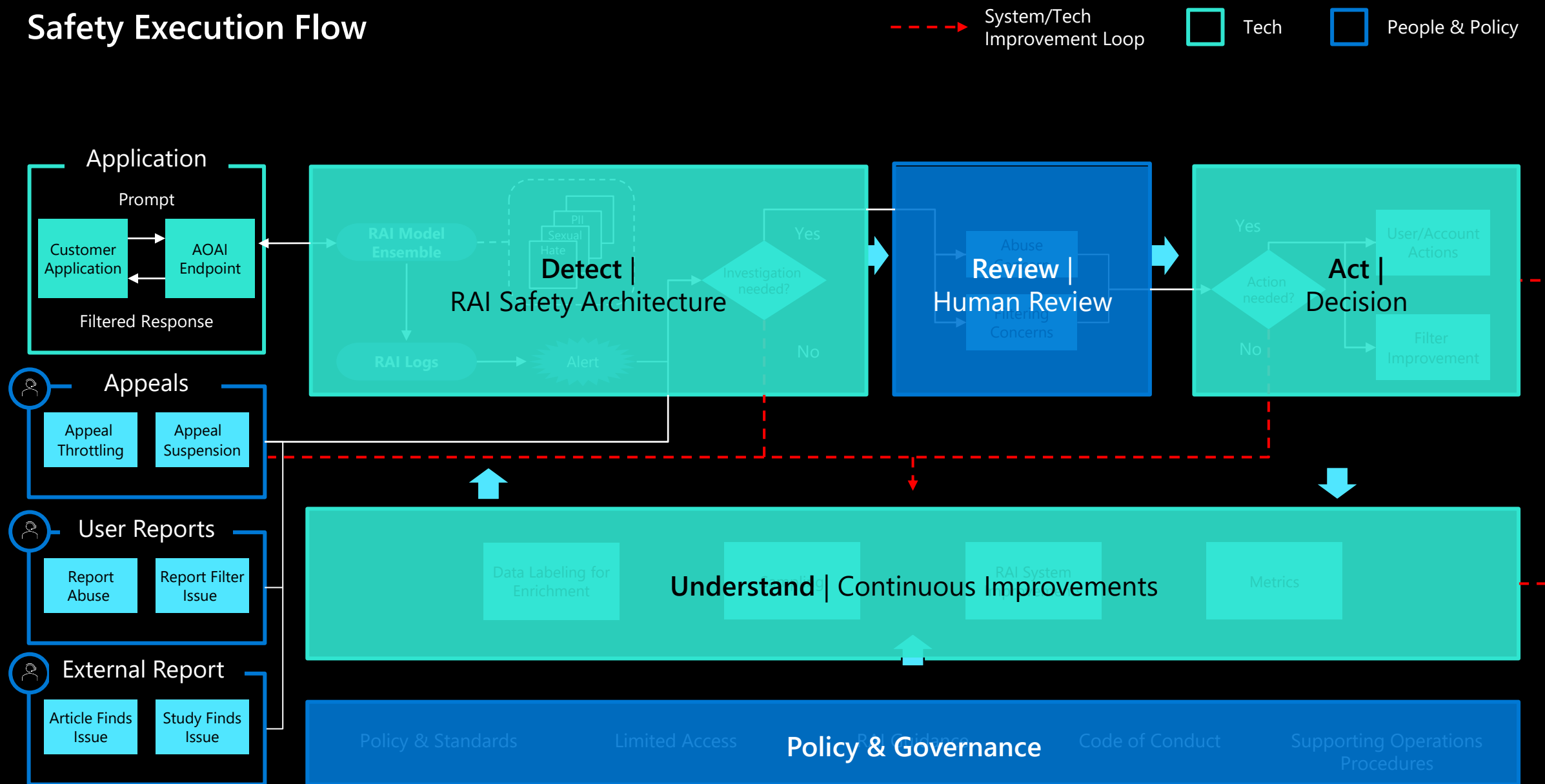
Content filtering—
can filter out abuse and misuse



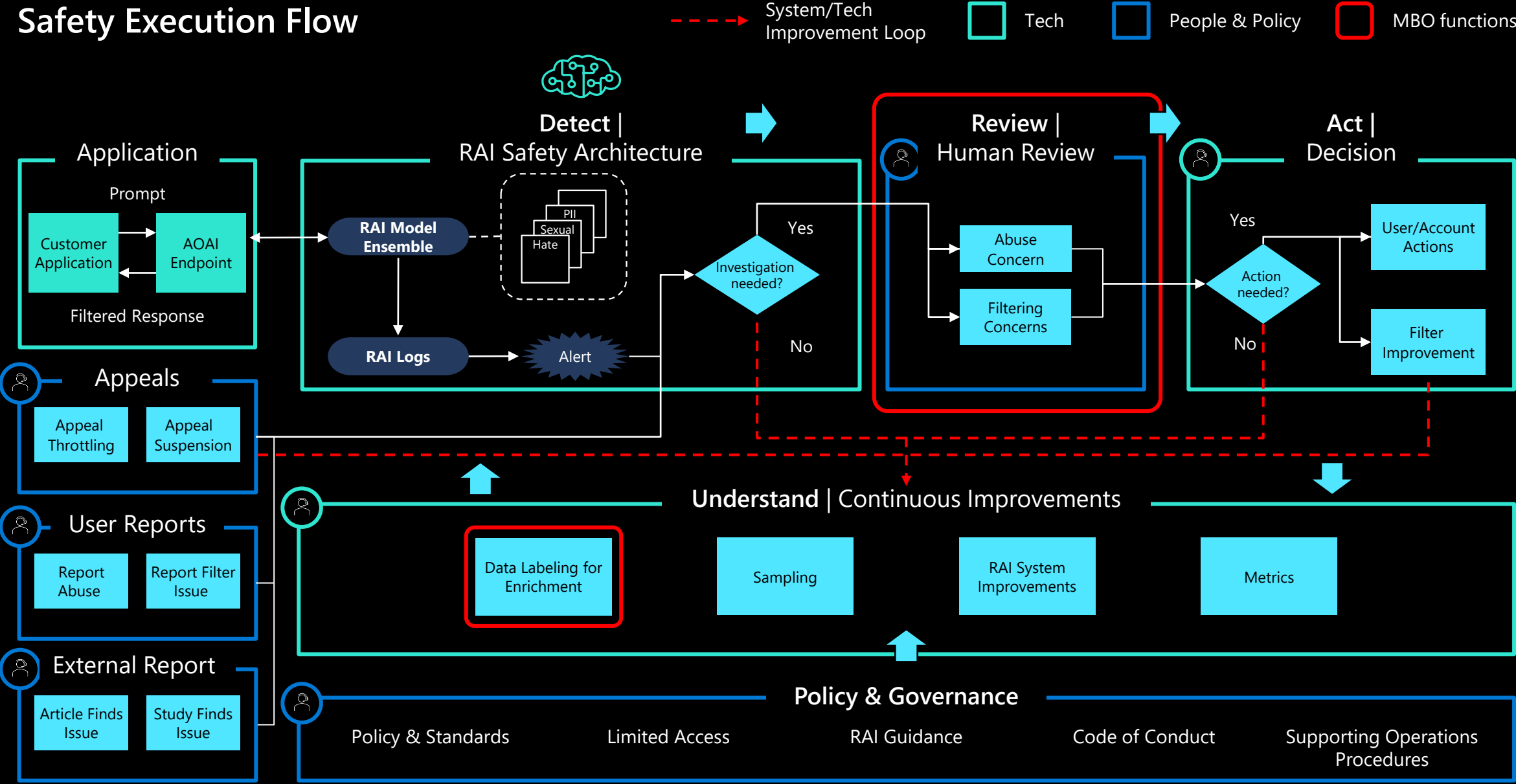
Implementation guidelines, patterns,
and best practices



Safety Execution Flow

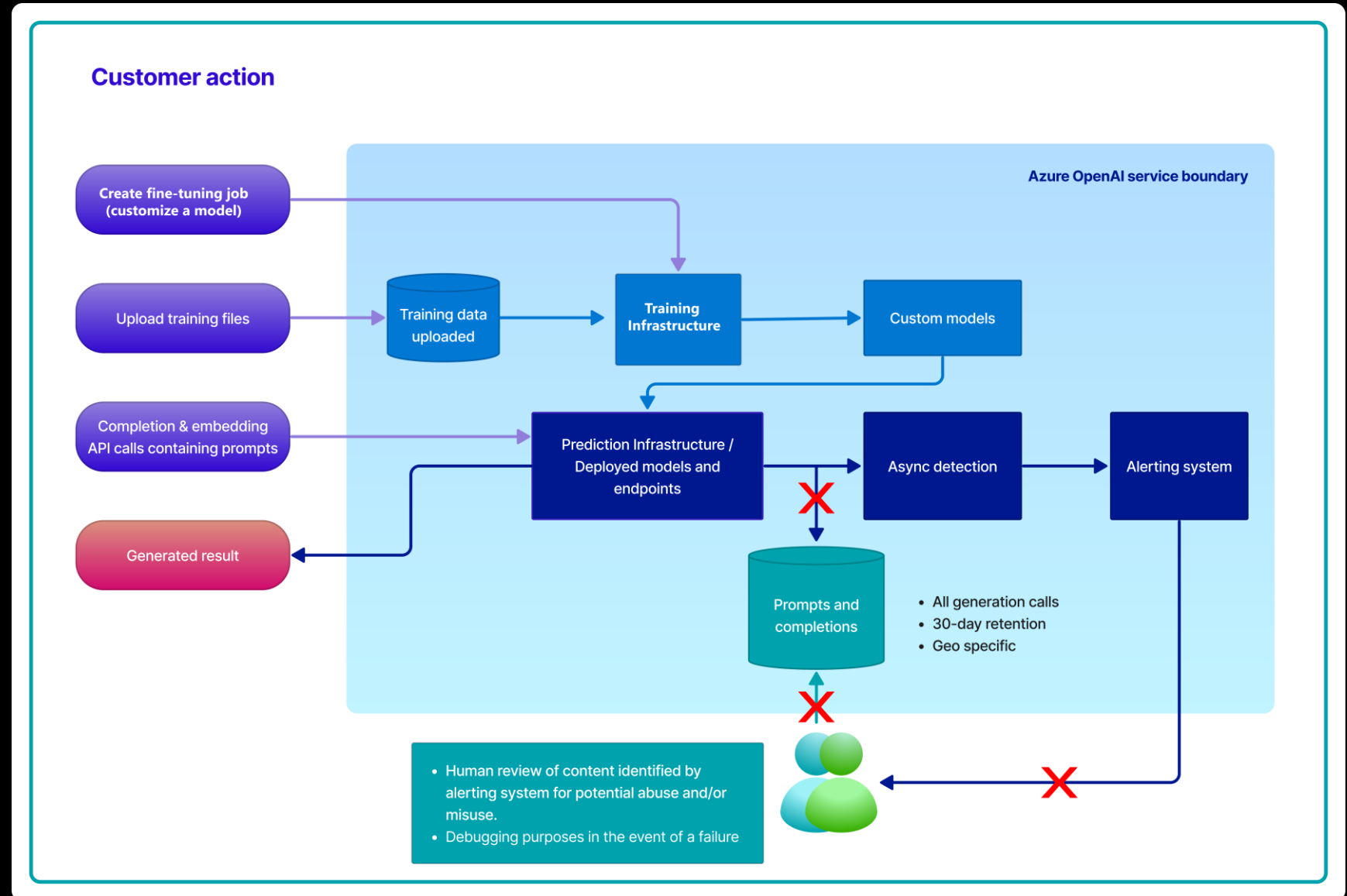


Safety Execution Flow

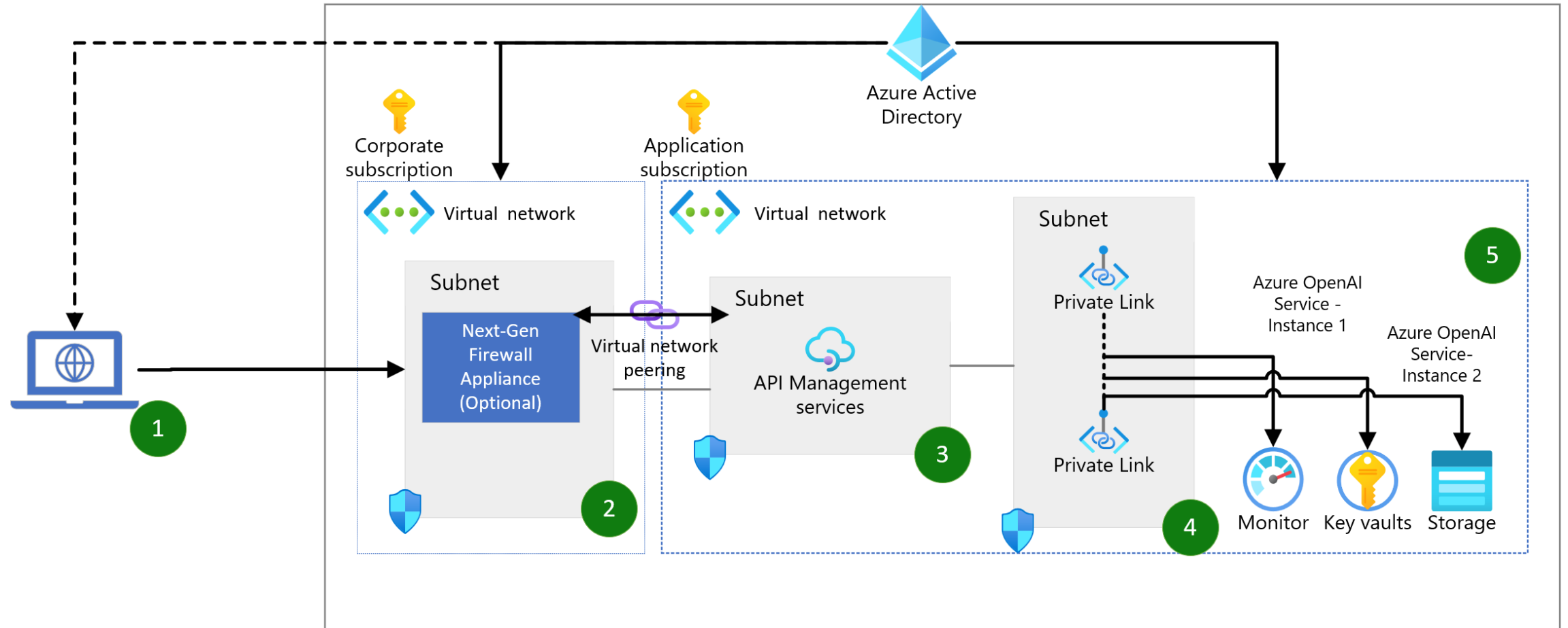


Data, privacy, and security for Azure OpenAI Service

<https://learn.microsoft.com/en-us/legal/cognitive-services/openai/data-privacy>



Enterprise Azure OpenAI



Responsible AI resources

Content filtering:

<https://learn.microsoft.com/en-us/azure/cognitive-services/openai/concepts/content-filter>

Responsible AI resources (aka.ms/RAIResources)

Human and AI Interaction Toolkit

(<https://www.microsoft.com/en-us/haxtoolkit/workbook/>)

Management tools

Evaluating and integrating Azure OpenAI for your use

Practices for responsible use

Ensure Human Oversight

- Let people edit generated outputs.
- Highlight potential inaccuracies in generated outputs.
- Remind users that they are accountable for final decisions and/or final content.
- Limit how people can automate your product or service.

Implement technical limits on inputs and outputs

- Limit the length of inputs and outputs.
- Structure inputs to limit open-ended responses and to give users more refined control.
- Return outputs from validated, reliable source materials.
- Implement blocklists and content moderation.
- Put rate limits in place.

Authenticate Users

- To make misuse more difficult, consider requiring that customers sign in and, if appropriate, link a valid payment method.
- Consider working only with known, trusted customers in the early stages of development.
- Applications that do not authenticate users may require other, stricter mitigations to ensure the application cannot be used beyond its intended purpose.

Test your application thoroughly

- Conduct adversarial testing where trusted testers attempt to find system failures, poor performance, or undesirable behaviors.
- Understand risks and consider appropriate mitigations.
- Communicate the capabilities and limitations to stakeholders.

Establish Feedback Channels for users and impacted groups

- Build feedback features into the user experience.
- Publish an easy-to-remember email address for public feedback.

Evaluating and integrating Azure OpenAI for your use

Scenario-specific practices

If your application powers chatbots or other conversational AI systems

Follow the Microsoft guidelines for [responsible development of conversational AI systems](#)

If you are developing an application in a high-stakes domain or industry

In healthcare, human resources, education, or the legal field, thoroughly assess how well the application works in your scenario, implement strong human oversight, thoroughly evaluate how well users understand the limitations of the application, and comply with all relevant laws.

Consider additional mitigations based on your scenario.

Learn more here [on our website](#)

API usage/rate limits

Requests per second per model	120 requests per minute* for: Davinci-models (002 and later) 300 requests per minute* for all other models.
Tokens per minute per model	40K tokens per minute* for: Davinci-models (002 and later) 120K tokens per minute* for all other models.

*all rate limits and availabilities are subject to change. Increased quotas can be requested, but not guaranteed.

[Learn more.](#)

Region availability

Regions currently available at launch:

- East US
- South Central US
- West Europe

To see the list of available regions, please visit:

[Product available by region](#)

REA	NORWAY	QATAR	SWEDEN	SWITZERLAND	UNITED ARAB EMIRATES	UNITED KINGDOM	UNITED STATES					
Products	tral	Sweden Central	Switzerland North	UAE North	UK South	UK West	Central US	East US	East US 2	North Central US	South Central US	
Azure Cognitive Services		✓	✓	✓	✓		✓	✓	✓	✓	✓	
Anomaly Detector		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Computer Vision		✓	✓	✓	✓	⬇	✓	✓	✓	✓	✓	
Content Moderator		✓	✓	✓	✓		✓	✓	✓	✓	✓	
Custom Vision					✓		⬇	✓	✓	✓	✓	
Face API		✓	✓	✓	✓		✓	✓	✓		✓	
Cognitive Service for Language			✓	✓	✓		✓	✓	✓	✓	✓	
Language Understanding (LUIS)			✓	✓	✓		✓	✓	✓	✓	✓	
Personalizer		✓	✓	✓	✓		✓	✓	✓	✓	✓	
QnA Maker			✓	✓	✓	✓	✓	✓	✓	✓	✓	
Speaker recognition												
Speech Services		✓	✓	✓	✓		✓	✓	✓	✓	✓	
Translator												
Azure OpenAI Service								✓			✓	

REST API | Reference

Operations

Delete	Deletes the file with the given file_id. Deletion is also allowed if a file was used, e.g., as training file in a fine-tune job.
Get	Gets details for a single file specified by the given file_id including status, size, purpose, etc.
Get File Content	Gets the content of the file specified by the given file_id. Files can be user uploaded content or generated by the service like result metrics of a fine-tune jo...
Import	Creates a new file entity by importing data from a provided url. Uploaded files can, for example, be used for training or evaluating fine-tuned models.
List	Gets a list of all files owned by the Azure OpenAI resource. These include user uploaded content like files with purpose "fine-tune" for training or validation o...
Upload	Creates a new file entity by uploading data from a local machine. Uploaded files can, for example, be used for training or evaluating fine-tuned models.

[Azure OpenAI REST API reference—Azure OpenAI | Microsoft Learn](#)

Azure OpenAI Studio

Getting started

Get started in the Studio

Refine your prompt

Copy your code

The screenshot shows the Azure OpenAI Studio interface. The top navigation bar includes 'Cognitive Services | Azure OpenAI Studio' and a user profile for 'Nick Brady'. A left sidebar contains navigation links: 'Azure OpenAI', 'Playground' (selected), 'Management', 'Deployments', 'Models', and 'File Management'. The main area is titled 'GPT-3 Playground' and features a 'Deployments' dropdown set to 'davinci-v2' and an 'Examples' dropdown set to 'Summarize Text'. A text input field contains a paragraph about neutron stars, with the cursor at the end. Below the input are 'Generate', 'Undo', and 'Regenerate' buttons, along with a 'Tokens: 144' indicator. On the right, a 'Parameters' panel includes sliders and input fields for 'Temperature' (0.7), 'Max length (tokens)' (60), 'Stop sequences', 'Top probabilities' (1), 'Frequency penalty' (0), 'Presence penalty' (0), 'Best of' (1), 'Pre-response text' (checked), and 'Post-response text' (checked). A 'Learn more' link is at the bottom right.

Pricing details

Please visit the Azure OpenAI Service [Pricing Details](#) page for the most up-to-date pricing information

What data does the Azure OpenAI Service process?

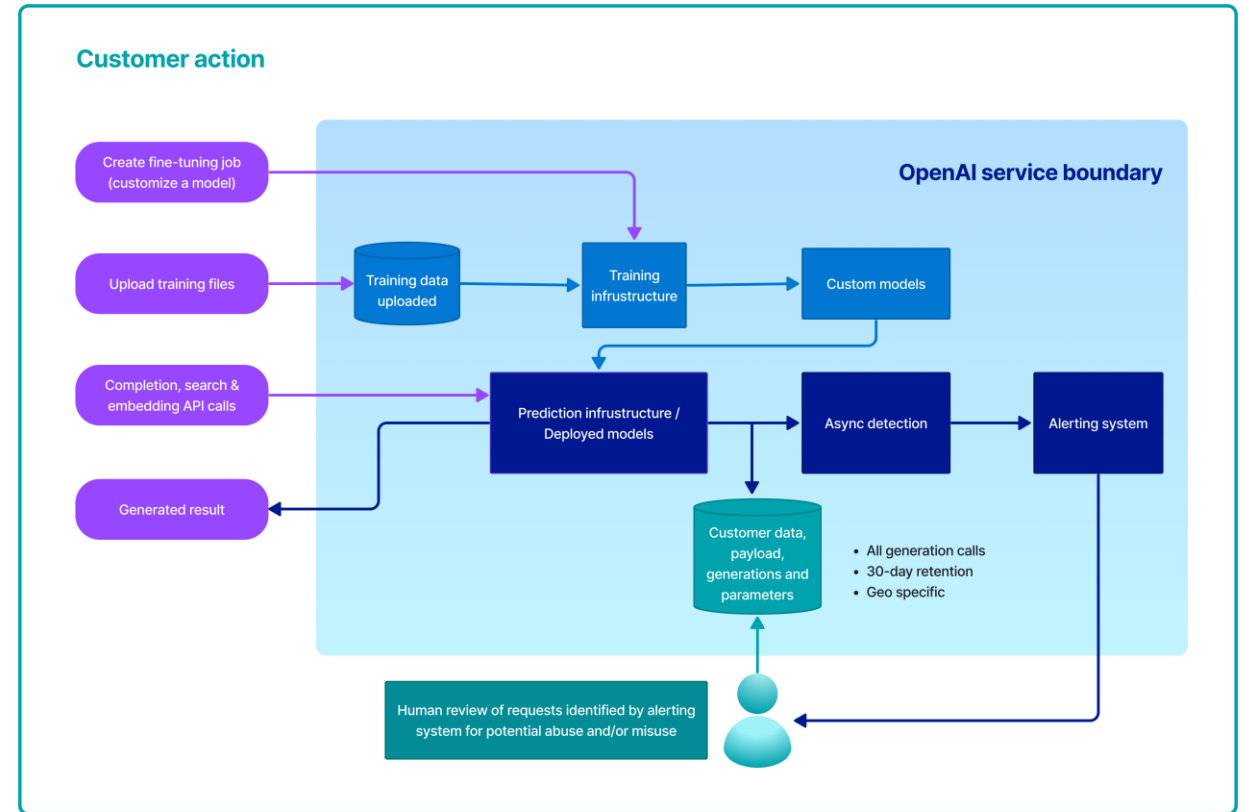
Azure OpenAI processes the following types of data:

Text prompts, queries, and responses submitted by the user via the completions, search, and embeddings operations.

Training and validation data. You can provide your own training data consisting of prompt-completion pairs for the purposes of fine-tuning an OpenAI model.

Results data from training process. After training a fine-tuned model, the service will output meta-data on the job which includes tokens processed and validation scores at each step.

How Azure OpenAI processes data



Training data to fine-tune an OpenAI model



The training data (prompt-completion pairs) submitted to the Fine-tunes API through the Azure OpenAI Studio is pre-processed using automated tools for quality checking including data format check.

The training data is then imported to the model training component on the Azure OpenAI platform.

During the training process, the training data are decomposed into batches and used to modify the weights of the OpenAI models.

Training data provided by the customer is only used to fine-tune the customer's model and is not used by Microsoft to train or improve any Microsoft models.

Text Prompts to generate completions, embeddings and search results

Once a model is deployed, you can generate text using this model using our Completions operation through the REST API, client libraries or Azure OpenAI Studio.



Abuse and harmful content generation

The Azure OpenAI Service stores prompts & completions from the service to monitor for abusive use and to develop and improve the quality of Azure OpenAI's content management systems. [Learn more about our content management and filtering.](#)

Authorized Microsoft employees can access your prompt & completion data that has triggered our automated systems for the purposes of investigating and verifying potential abuse; for customers who have deployed Azure OpenAI Service in the European Union, the authorized Microsoft employees will be located in the European Union.

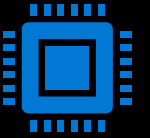
This data may be used to improve our content management systems.

In the event of a confirmed policy violation, we may ask you to take immediate action to remediate the issue to and to prevent further abuse.

Failure to address the issue may result in suspension or termination of Azure OpenAI resource access.

A customer can opt out of logging and human review process—learn more [here](#).

Training, validation, and training results data

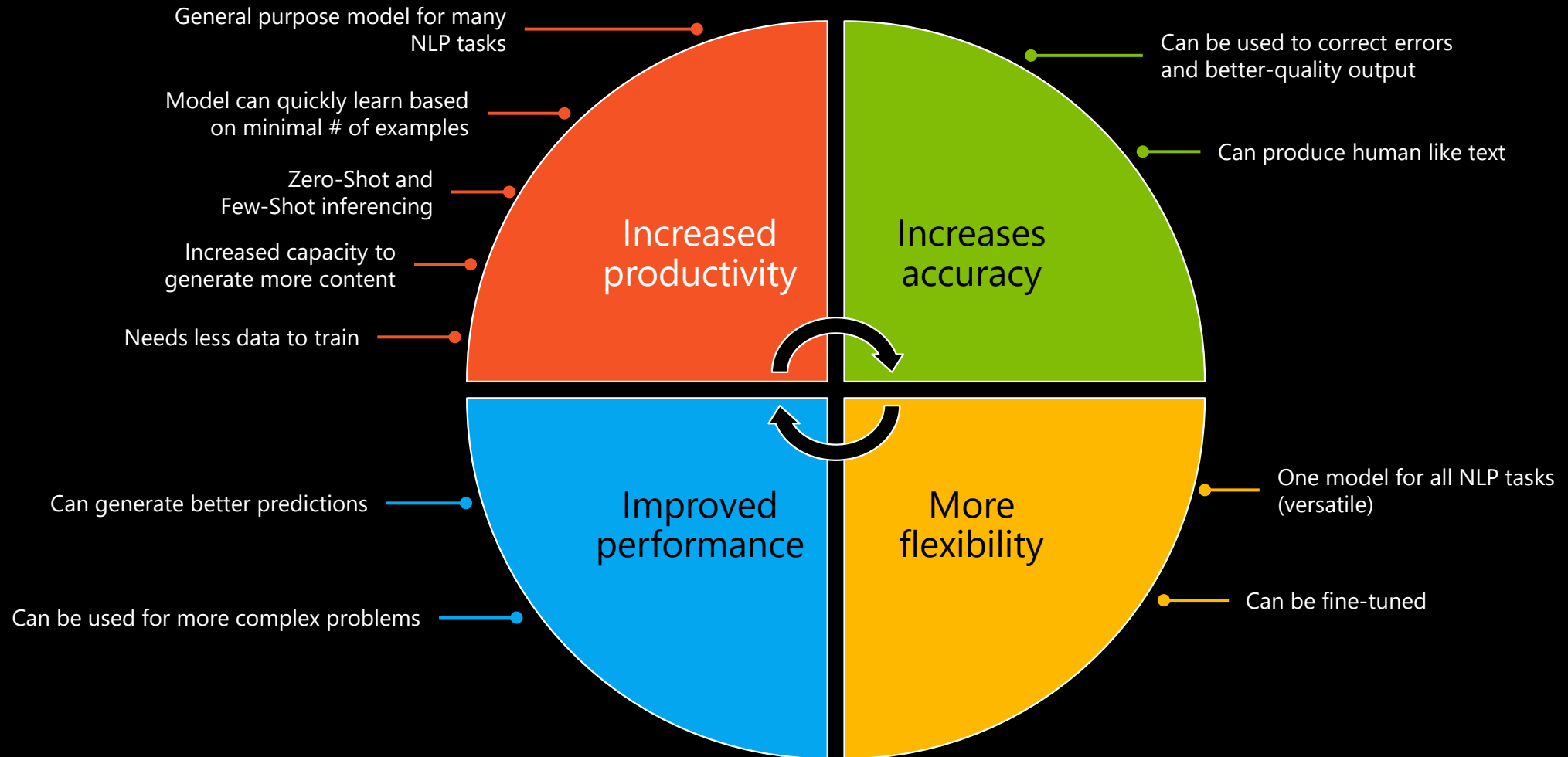


The Files API allows customers to upload their training data for the purpose of fine-tuning a model.

This data is stored in Azure Storage, encrypted at rest by Microsoft Managed keys, within the same region as the resource and logically isolated with their Azure subscription and API Credentials.

Uploaded files can be deleted by the user via the DELETE API operation.

Why use GPT-3?



Disadvantages of GPT-3 Models

- Lacks interpretability and explainability
- Limited input size
- More computationally expensive
- Risk of hallucinations
- Can have bias





The margins in construction are narrow, so it's important for companies to pick the right projects to work on.

Strabag SE used Azure OpenAI Service to build an AI-based risk management solution that assesses which construction projects might fail – therefore minimizing the risks that come with it and saving time and money.

"With just three months of data, this algorithm can predict risk with 80% accuracy – it's mind-blowing!"

- Dr. Marco Bornschlegl, Head of Innovation and Digitalisation

Strabag SE





Since 1993, CarMax has grown from an industry-disrupting startup to become the United States' leading used car retailer.

CarMax uses Azure OpenAI Service to streamline the creation of text summaries for its car research pages, quickly providing customers with meaningful content that also boosts the pages' search engine rankings.

"With the help of OpenAI Service, we're disrupting our industry for a second time by delivering cutting-edge digital tools and capabilities and becoming a true omnichannel retailer."

Shamim Mohammad, Executive Vice President and Chief Information and Technology Officer

CarMax





Trelent uses cutting-edge AI algorithms to provide intelligent source code documentation for its clients.

To scale its solution and accommodate the data security and residency needs of enterprise customers around the globe, Trelent chose to migrate its solution to Microsoft Azure OpenAI Service.

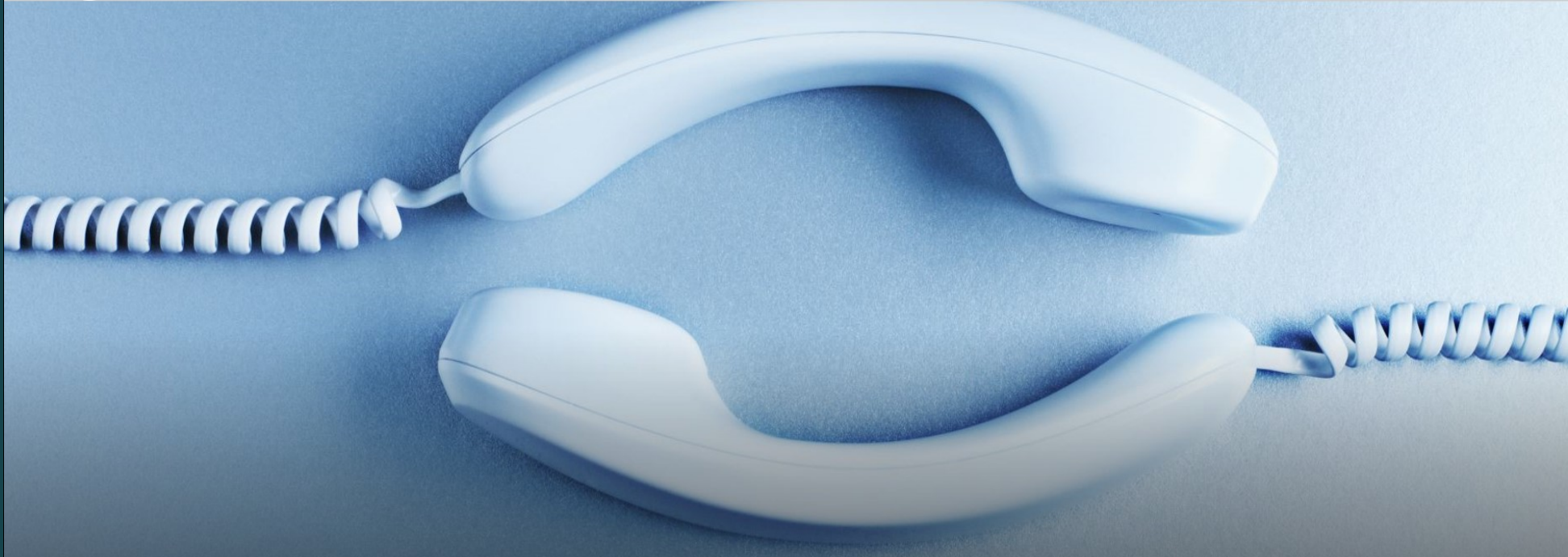
"With a product like OpenAI Service behind you, you can focus a lot more on what really matters, which is delivering a great experience, a great product, and a lot of value to your customers."

Calum Bird, CEO

Trelent



Customer name masked



- CUSTOMER:**
- Large Telco Organisation
- INDUSTRY:**
- Telco
- SIZE:**
- Corporate (10,000+ employees)
- REGION:**
- Europe
- PRODUCTS AND SERVICES:**
- Azure OpenAI Service : Call Summarisation

Situation:

This company currently has a virtual agent that fields inquiries, usually with little to no human intervention. If the inquiry is beyond the bot's comprehension, it will hand over the conversation to human agents. Even with the bot adept to handle the initial queries, there is still significant manual effort by human agents to figure out the scope of the problem, and for them to be able to quickly detect new problems.

Proposed:

In order to reduce the time for handoffs to the human agents and to help understand the problem faster and more efficiently, the company is using Azure OpenAI Service to help summarize the conversation in a few sentences.

The customer is also using Azure OpenAI Service for call summarization to detect new problems.

Areas of Impact:

- Time savings for agent, and summaries are used to detect new topics in the call center to better react to new problems.
- Cost Savings.

Customer name
masked



CUSTOMER:

- Large Automotive Organisation

INDUSTRY:

- Automotive

SIZE:

- Corporate (10,000+ employees)

REGION:

- Americas

PRODUCTS AND SERVICES:

- Azure OpenAI Service : Content Generation

Situation:

This company currently spends a fair amount of time in manually generating content to help customers.

Proposed:

In order to automatically generate content in a self-help manner to their customers, this company is using Azure OpenAI's content generation feature to reduce the amount of manual effort required in responding to their customers.

The solution would be in an conversational manner, and integrated with their existing applications / websites.

Areas of Impact:

- Time savings
- Cost savings
- Productivity increasing
- Revenue increasing through hyper-personalisation

Customer name
masked



CUSTOMER:

- Online Game Platform

INDUSTRY:

- Gaming / Technology

SIZE:

- ISV (1000+ employees)

REGION:

- Americas

PRODUCTS AND SERVICES:

- Azure OpenAI Service :
Content Generation

Situation:

This company provides an interactive gaming experience on their platform and they want to be able to quickly and innovatively develop code on their platform.

Proposed:

This company connected with the Azure OpenAI Service team to develop a code completion service within their IDE, taking a co-pilot approach to developing code on the platform. Through the power of Codex, they can meet developers where they are as they bring interactive gaming experiences to life on the platform

Areas of Impact:

- Time savings
- Innovation
- Revenue increasing

Customer name
masked



CUSTOMER:

- Large Oil & Gas Company

INDUSTRY:

- Oil & Gas / Energy

SIZE:

- Corporate (10,000+ employees)

REGION:

- Americas

PRODUCTS AND SERVICES:

- Azure OpenAI Service :
Question & Answer

Situation:

This company has been working with Microsoft to develop chatbots for internal and external interaction and Cognitive Search for searching through large documentation.

Proposed:

The company wants to use Azure OpenAI Service to train their chatbots on being able to handle more controversial topics.

The company would extract and summarise key information from documentation and other channel for internal employees to be able to consume the outputs in a Q&A format.

Areas of Impact:

- Time Savings
- More relevant and all-round information
- Employee satisfaction through more succinct information.

Customer name
masked



- CUSTOMER:**
- Large Mass Media Company
- INDUSTRY:**
- Mass Media
- SIZE:**
- Corporate (10,000+ employees)
- REGION:**
- Americas
- PRODUCTS AND SERVICES:**
- Azure OpenAI Service : Content Generation

Situation:

This company is a leading media company who wants to accelerate and analyse all pre-publication work of journalists. They also have a lot of video content that needs to be extracted and analysed.

Proposed:

The company is planning to use Azure OpenAI Service for content edits to comply with their editorial style guides to generate compelling summaries generating SEO optimized headlines for articles, assisting in article composition from wires or from fact lists, etc.

They also want to perform entity extraction and generate summaries of audio transcripts from video content.

Areas of Impact:

- Time savings
- Innovative content generation

APPENDIX

Resources to learn more

[Microsoft AI & ML University](#)

FAQ

Responsible AI

- Microsoft has filters on content in the OpenAI service to align our Responsible AI principals. Can we get details on how these filters are implemented, monitored, and enforced?
- If a principal is violated, how is that flagged and what recourse is taken?

Papers

Links

Attention is all you need (Transformers)—

[\[1706.03762\] Attention Is All You Need \(arxiv.org\)](#)

Proximal Policy Optimization (PPO)—

[\[1707.06347\] Proximal Policy Optimization Algorithms \(arxiv.org\)](#)